

Cybercrime

Lecture Notes & Course Readings

M.Sc. in Artificial Intelligence

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Spring Semester 2025–2026

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PART I

Foundations of Cybercrime

CHAPTER 1

Crime in the Digital Society

1.1 Digitalisation: Definition and Short History

1.1.1 Digital Data

Definition 1.1 (Digital data). Electronic representation of information in a format or language that computers can read and understand. In more technical terms: information in binary format (0–1) that is converted into a digital format readable by a computer.

1.1.2 A Brief History of Digitalisation

- **1663:** John Graunt applies statistical analysis to mortality data from the bubonic plague — an early precursor to data-driven reasoning.
- **1865:** R. Millar Devens coins the concept of business intelligence.
- **1884:** Herman Hollerith builds the tabulating machine for the US Census; later founds what becomes IBM.
- **1943:** Colossus, the first programmable electronic computer, is used to decipher Nazi communications at Bletchley Park.
- **1989:** Tim Berners-Lee creates the World Wide Web.
- **1995:** First supercomputer.

1.1.3 From Digital Data to Big Data

- **Since 1997:** Emergence of social networks.
- **1999:** The term **Internet of Things (IoT)** enters use — the extension of the Internet to the world of physical objects.
- **2005:** Roger Mougalias coins the term **Big Data**: a set of digital data so large that it cannot be analysed and stored using traditional tools; specific technologies and analytical methods are needed to extract value from it.

1.2 Case Study: Smart Toys and Criminal Risks

This case study illustrates how IoT consumer devices — specifically smart toys — introduce concrete criminal risks by collecting and transmitting sensitive data about children.

1.2.1 The Cayla Doll

The Cayla doll was an internet-connected toy capable of recording and transmitting children's voice data. In 2017, Germany banned the device over fears that hackers could exploit it to target children. The case raises a fundamental question that recurs throughout the course: *unfounded fears or documented reality?*

1.2.2 CloudPets

CloudPets were internet-connected stuffed animals that allowed children and parents to exchange voice messages via a companion app. A security analysis by Troy Hunt revealed that the backend database — containing millions of voice recordings and user credentials — was left exposed on the internet without authentication, was subsequently accessed by unauthorised parties, and the data was held for ransom. This is a documented case of a data breach compounded by ransom, affecting a particularly vulnerable population.

1.2.3 Wiggy

Wiggy was a crowdfunded “smart piggy bank” that linked to a child’s savings account via a mobile app, allowing parents and relatives to deposit money remotely. The device combined financial account access with an IoT object marketed at children, creating a direct bridge between physical play and banking infrastructure — a surface area for financial fraud and account compromise.

1.3 Crime in the Digital Society

The digital environment does not create crime *ex novo*, but it transforms its scale, speed, authorship, and victim profile. The following dimensions illustrate this transformation through concrete examples.

1.3.1 Crime Types

Digital infrastructure enables new crime typologies or drastically scales existing ones:

- **Cyberattacks on critical infrastructure:** In February 2023, a pro-Russian hacker group targeted Italian government websites and corporate portals (Carabinieri, Bper bank, A2A, TIM) coinciding with Prime Minister Meloni’s visit to Kyiv. The Italian National Cybersecurity Agency (ACN) managed to mitigate the effects.
- **Election manipulation:** Investigative reporting identified a network led by “Tal Hanan” (former Israeli special forces operative) that allegedly manipulated 30 elections worldwide by deploying hacking, disinformation, and social engineering campaigns on behalf of paying clients.
- **Online fraud and sextortion:** Italian police data and news reports document a continuous rise in online fraud, with sextortion and fake social profiles representing a significant and growing share.
- **AI-assisted fraud:** In February 2025, a fraud scheme used AI-generated voice cloning to impersonate Italian Defence Minister Crosetto and a cloned ministerial phone number to solicit money from major Italian entrepreneurs (Armani, Moratti, Tronchetti Provera, Della Valle) under the pretext of “freeing hostages”. At least one victim paid.

1.3.2 Crime Authors

The digital environment radically diversifies who can commit crime and at what scale:

- **Individual actors:** Cody Wilson (b. 1988), founder of Defense Distributed, published open-source gun designs (“wiki weapons”) for 3D printing online, gaining international notoriety in 2013 with the Liberator — the first widely available 3D-printed pistol.
- **Teenage actors:** Graham Ivan Clark, at age 17, hacked Twitter’s internal admin tool in 2020 and took control of high-profile accounts (Obama, Gates, Musk, West) to run a

Bitcoin scam that defrauded victims of over \$100,000. He pleaded guilty to a three-year sentence.

- **Nation-state actors:** Bureau 121, North Korea's internal hacking unit, has been linked to several major malware events, most notably the **WannaCry ransomware attack** (2017), which infected around 300,000 devices across 150 countries and caused an estimated \$4 billion in damages. The US sanctioned North Korea over WannaCry in 2019.
- **AI systems as unintended actors:** Microsoft's Tay chatbot (2016) was corrupted by coordinated users within 24 hours of launch and began producing racist and misanthropic content at scale — a case where the crime author is ambiguous and the platform's own technology becomes the vector.

1.3.3 Crime Victims

Victimisation in the digital society is broader, faster, and less visible than in traditional crime:

- **Sextortion:** FBI data documents a troubling increase in sextortion cases targeting minors. In one case (May 2022), a 17-year-old died by suicide hours after being scammed.
- **Mass data breach:** The Ashley Madison hack exposed the personal data of 32 million users of an extramarital affairs platform, with ongoing blackmail and reputational damage for victims years after the breach.
- **Cyber kidnapping:** A new criminal trend targeting foreign exchange students and their families. Police found a Chinese exchange student isolated and frightened in a tent in the Utah mountains; the perpetrators had convinced the victim to stage a kidnapping while extorting the family remotely.
- **Digital-enabled physical crime:** Thieves monitored Facebook posts to identify when homes were empty before burglarising them. In the Kim Kardashian Paris hotel robbery (2016), the perpetrators reportedly used her social media to track her location and movements.
- **Digital humiliation:** In Milan (2023), a perpetrator filmed and shared on social media footage of a physical assault she committed, leading victims' counsel to demand a public apology on TikTok as part of the settlement.

1.3.4 Social Control and Policing

Digital technologies also reshape the instruments of social control:

- **Facial recognition:** In 2020, the European Commission considered a five-year ban on facial recognition in public spaces, highlighting the tension between law enforcement utility and civil liberties.
- **Algorithmic risk assessment:** Police forces increasingly use algorithmic tools to assess threat levels and predict criminal behaviour. A 2016 ProPublica investigation ("Machine Bias") documented that the COMPAS recidivism prediction software used across US courts exhibited systematic racial bias, falsely flagging Black defendants as future criminals at nearly twice the rate of white defendants.

A recurring tension throughout the course: the same digital tools that enable crime also constitute the primary instruments of crime detection and social control. This duality — and its legal, ethical, and criminological implications — is the central analytical lens of the course.

CHAPTER 2

Examples of Criminological Studies

2.1 Confessions of a Migrant Smuggler

An ethnographic study by Andrea Di Nicola and Giampaolo Musumeci into migrant smuggling as a classic organised crime activity with first elements of digitalisation. The research was conducted in the years before the Mediterranean migration crisis and consisted of a two-year journey inside the minds of human smugglers, presenting the downside of the coin and debunking stereotypical ideas about “scafisti”.

2.1.1 First Elements of Digitalisation

Smugglers were found to use technology and digital tools for several purposes:

- For organisational purposes and to conduct their activities.
- To communicate with other smugglers in the network.
- To recruit clients and market their services.
- To acquire the tools necessary for the business (e.g. falsified documents, credit card information).

New means of communication adopted by smuggling networks include Skype, satellite phones and cellphones, wireless networks, email, and instant messaging apps.

2.2 Project Surf and Sound: The Digitalisation of Migrant Smuggling

The Surf and Sound project aimed at “improving and sharing knowledge on the Internet’s role in the human trafficking process”. It was the first research project to produce systematic knowledge on the use of the Internet (including social networks) in trafficking in human beings (THB) and migrant smuggling, disseminating findings through workshops for law enforcement agencies (LEAs) and NGOs. The project was co-financed by the European Commission — DG Home Affairs within the ISEC 2013 “Prevention and Fight against Crime – Trafficking in Human Beings” programme.

2.2.1 Methodology

The study employed a mixed-methods design:

- **Internet exploration (virtual ethnography):** passive and active virtual ethnography ($N = 939$).
- **In-depth interviews** ($N = 58$): LEAs (16), NGOs (13), victims/clients (14), traffickers/smugglers (13), and cybercrime experts (2).
- **Online web survey** ($N = 995$): analysing the experience of internet users with web content linked to human trafficking and people smuggling.

2.2.2 Key Results

1. The Internet plays a relevant role in the smuggling of migrants.
2. Content linked to migrant smuggling services on social media websites is easily accessible.
3. Social media websites facilitate the advertisement of smuggling services; Facebook and Instagram are more used than Twitter.
4. Social media pages and profiles run by smugglers provide information on travels.
5. Social media profiles and pages run by smugglers are very effective and responsive.
6. Smugglers provide potential migrants with detailed information via mobile applications.
7. Social media websites play a key role in the decision-making process of potential migrants.
8. Social media websites are used by migrants to post feedback about smuggling services.
9. The Internet is used also during and after the journey.
10. The deep/dark web seems to play a marginal role.

2.3 Project FAKECARE: The Digitalisation of the Illegal Trade of Medicines

2.3.1 The Problem

An estimated **10%** of medicines worldwide are fake, illegal, or substandard. There are significant geographical differences: in developed countries the figure is below 1%, while in developing countries it ranges from 15–30% (sometimes reaching 50%). This results in an estimated 100,000 to 700,000 deaths annually, with a global estimated turnover of 75 billion USD per year (**10 billion EUR/year in the EU**).

2.3.2 Falsified Medicinal Products

The use of several terms (“counterfeit”, “falsified”, “false”, “tampered”) creates great confusion. For clarity, falsified medicinal products are **medicines that are dangerous** for private and public health. The umbrella category of **illegally traded (or produced) medicines** (ITMs) includes:

- Falsified branded medicines
- Falsified generic medicines
- Illegal medicines
- Medicines without authorisation
- Medicines with expired authorisation
- Legal (and genuine) medicines sold without a valid prescription

2.3.3 Methodology

Quantitative methods.

- **Web survey:** profiling the risk for customers to buy a falsified medicinal product online.
- **Honey-pot websites:** two fake online pharmacy websites set up to attract customers, automatically monitor their behaviour, and calculate statistics on visitors, clicks, and checkouts.

- **Quantitative content analysis:** on online pharmacies, customers' reviews on illegal online pharmacies, and news from the press.
- **Statistics** on the online trade of falsified medicines using data provided by regulatory agencies, law enforcement agencies, and Interpol.

Qualitative methods.

- Analysis of **EU regulations** in the field of the online trade of falsified medicines.
- **Virtual ethnography:** exploring online forums and communities (including social networks).
- **Crime script analysis:** analysing investigative and judicial cases, news from the press.
- **Critical quantitative analysis of secondary sources.**

2.3.4 Crime Script of the Online Trade

The online trade can be seen as a process rather than a single event, involving stages in which resources and locations are required and decisions are made. The focus is mainly on the **supply side**:

1. **Preparation:** identification of the criminal opportunities to commit the online trade in falsified medicinal products.
2. **Pre-activity:** logistical or transactional instrumental actions prior to the activity.
3. **Activity:** concrete actions necessary to carry out the trade, which can take the form of various offences of non-compliance.
4. **Post-activity:** logistical or transactional activities necessary to leave the scene.

2.3.5 The Role of the Internet in Distribution

There are significant geographical differences in distribution channels:

- **Developing countries:** legal distribution chain.
- **Developed countries:** illegal distribution chain (illegal websites, online marketplaces, social media websites, etc.).

An estimated **94%** of online pharmacies (around 34,000) operate illegally.

2.3.6 Indicators of an Illegal Online Pharmacy

- No request of a prescription for prescription-only medicines
- Low prices, promotions, and offer of trial packs
- Use of different languages, grammar mistakes
- Absence of a physical address
- Promotion of medicinal products on the home page
- Great attention to providing anonymity
- Use of "testimonials"

2.3.7 Distribution Channels

Falsified medicines are distributed through multiple channels: social media platforms (Facebook, Twitter, Instagram), online marketplaces (e.g. Alibaba), and the dark web (e.g. the former Silk

Road marketplace).

2.3.8 EU Common Logo for Online Sale of Medicines

As of 1 July 2015, all legally operating online pharmacies/retailers in the EU must display a common logo. The logo must be clearly displayed on every page of the website and links to the website of the national competent authority and to the national list of legally operating online pharmacies/retailers.

However, the system is **vulnerable to fraudsters**. A vulnerability assessment demonstrated that the EU common logo and the verification scheme can be counterfeited in a few minutes, with basic computer skills, spending only the money necessary to purchase a fraudulent domain. The steps are:

1. Copy the logo from a legitimate website.
2. Copy the institutional webpage of the national authority.
3. Falsify the institutional webpage to list the illegal pharmacy.
4. Purchase a fraudulent domain with a similar URL.

In addition, the verification scheme has not been well implemented in some cases: the logo is not displayed at all, or the link does not work or points to pages other than the national register.

CHAPTER 3

Cybercrime Typologies: Cyber-Dependent Crimes

3.1 Categories of Cybercrime

The term **cybercrime** covers a wide range of deviant and criminal behaviours that are constantly changing and evolving. It is unlikely that there is an exhaustive list or universally accepted definition of cybercrime. More common definitions refer to categorisations of cybercrime (Phillips et al., 2022), and these categories are not rigid — there is the possibility of partial overlap. Cybercrime is often defined using **dichotomies and trichotomies**.

3.1.1 The Brenner–Wall Trichotomy

Brenner (2007) proposed a dichotomy later extended by Wall (2010) into a trichotomy:

- **Cyber-dependent crime:** crimes that arose with technology and cannot exist outside the digital world.
- **Cyber-enabled crime:** the process of digitalisation has facilitated the commission of these crimes.
- **Cyber-assisted crime:** traditional crimes in which the use of the computer is not essential but rather incidental.

3.1.2 The Gordon–Ford and Sarre–Lau–Chang Trichotomy

Gordon and Ford (2006) distinguished:

- **Type I:** technical cybercrimes.
- **Type II:** cybercrimes that involve human contact.

Sarre, Lau, and Chang (2018) added a third category:

- **Type III:** crimes perpetuated by AI, robots, bots, and self-learning technology.

3.2 Cyber-Dependent Crimes

Definition 3.1 (Cyber-dependent crime). High-tech crimes that can only be committed using computers, computer networks, or other forms of ICT (Wall, 2005; Europol, 2017). These are crimes that **can only be committed in cyberspace** and would not exist without the Internet or technology. They represent a new form of crime that has emerged with the digital society — also referred to as *true cyber crime* or *real cyber crime*.

3.2.1 Subcategories by Mode of Execution

Based on how they are carried out, cyber-dependent crimes can be divided into **two subcategories** (McGuire & Dowling, 2013):

1. Illicit intrusions into computer networks:

- Hacking
- Unauthorised access to computer networks, computers, and mobile devices exploiting security gaps and vulnerabilities in order to collect personal data and useful information, or to deface websites and launch DoS/DDoS attacks

2. Interruption or damage to the functioning of computers and networks:

- Malware
- Spam
- DoS
- DDoS
- Botnets

3.2.2 Subcategories by CIA Triad

Brar and Kumar (2018) proposed a **threefold definition** based on the pillars of cybersecurity:

- **Attacks on confidentiality:** eavesdropping, traffic analysis, password attacks, snooping, keyloggers, social engineering.
- **Attacks on integrity:** salami attacks, data diddling, cross-site scripting (XSS), SQL injection, session hijacking.
- **Attacks on availability:** DoS, DDoS.

3.3 Case Studies

3.3.1 Ransomware Attacks on Hospitals

On 3 August 2023, a ransomware attack hit a California-based healthcare system operating 16 hospitals and more than 165 clinics and centres across four US states. As a result, several services were shut down, facilities closed, patients and ambulances diverted, and digital records were unavailable. Cyberattacks on hospitals are on the rise worldwide (Argaw, Bempong, Eshaya-Chauvin et al., 2019).

3.3.2 Ransomware Attacks on Law Enforcement

In August 2023, a data breach at a company that prints ID cards and lanyards compromised data from customers including Greater Manchester Police and the London Metropolitan Police. The exposed data included names, ranks, photos, and serial numbers. Europol conducted an operation that led to the arrest of a member of the Ragnar Locker group and the seizure of its infrastructure.

3.3.3 DDoS Attacks

DDoS attacks are one of the biggest cyber threats and have evolved to avoid detection and increase damage (Bergamini de Neira, Kantarci, Nogueira, 2023). In October 2023, a record-breaking DDoS attack was reported against Google, Cloudflare, and AWS, peaking at over 398 million requests per second (RPS). Amazon (June 2020) and Google (2022) had previously reported stopping the then-largest DDoS attacks. The scale has grown from 26 to 398 million requests per second.

3.3.4 Ashley Madison

In July 2015, a hacker collective called “The Impact Team” infiltrated Ashley Madison — a dating website for those seeking extramarital affairs — stealing over 60 gigabytes of sensitive data (user profiles, financial transactions, sexual orientations, internal communications). Personal information of approximately 32 million users was exposed, leading to personal crises, suicides, and allegations of fraud and deceptive business practices by the platform.

3.3.5 The Role of Artificial Intelligence

AI introduces new dimensions to cyber-dependent crime:

- **Acoustic side-channel attacks:** researchers developed deep-learning systems capable of hacking passwords using key acoustics with over 90% accuracy (Harrison, Toreini, Mehrnezhad).
- **AI-enhanced social engineering:** chatbots such as ChatGPT can be exploited to create convincing phishing emails by correcting spelling and grammatical errors that would otherwise alert human readers. “ChatGPT may therefore provide criminals with new opportunities, particularly for crimes involving social engineering” (Europol, 2023).

CHAPTER 4

Cybercrime Typologies: Cyber-enabled Crimes

4.1 Introduction and Definition

The digital society has given a **strong boost to a number of traditional crimes** that have been enhanced by the use of computers, computer networks, or other forms of information and communication technology (ICT). In contrast to cyber-dependent crime, **cyber-enabled crime** (McGuire & Dowling, 2013) **can also be committed without the use of the digital world**: it is not born with technology, but technology transforms its scale, reach, and ease of commission.

A key distinction: cyber-enabled crime *has always existed*, but it is now more **ubiquitous, faster**, and more **global**. The internet does not create new criminal motivations, but it dramatically expands opportunities and reduces barriers.

Definition 4.1 (Cyber-enabled crime). Traditional crimes for which the Internet has opened up entirely new possibilities and opportunities; crimes that are facilitated or have been made easier (i.e., enabled) by cyber technology (Phillips et al., 2022). These are crimes where the computer plays a role but that could still be committed without the involvement of the computer. Cyber-enabled crimes range from white-collar crime to drug trafficking, to online harassment, terrorism, and beyond.

Also referred to as:

- **Hybrid crimes** (Wall, 2005)
- **Computer-related crimes** (Broadhead et al., 2018)
- **People crimes** (Black et al., 2019)

Remark. The boundary between cyber-dependent and cyber-enabled crime is not always clear-cut. The categories are analytically useful but descriptively fluid: the same conduct can shift categories depending on the specific *modus operandi* employed.

4.2 Subcategories of Cyber-enabled Crime

Cyber-enabled crimes are organised into two broad clusters based on the nature of the harm caused:

4.2.1 Cyber-enabled Crimes Related to Economy

This cluster includes offences in which the primary motive is financial gain and the digital environment provides the infrastructure, anonymity, and scale needed to commit them:

- Fraud (e-commerce fraud, mass-marketing frauds, phishing scams, pharming scams, “online romance” frauds)
- Intellectual property crimes

- Sale of illegal items on online marketplaces (including the deep or dark web)
- Cyber-laundering

4.2.2 Cyber-enabled Crimes Related to Interpersonal Violence

This cluster encompasses offences targeting individuals, often with psychological, reputational, or sexual harm as the primary outcome:

- Cyber-bullying
- Cyber-mobbing
- Cyber-stalking
- Cyberharassment
- Sextortion
- Online grooming
- Trolling
- Griefing
- Assisted cyber suicide
- Image-based abuse (IBA)
- Production and dissemination of child pornography

4.3 Cyber-enabled Crimes Related to Economy

4.3.1 The Industrialisation of Fraud

The world of fraud is one of the areas that has benefited most from the introduction of the internet. The digital society has **industrialised fraud**: it lowered the barriers to entry, allowed offenders to reach exponentially more victims simultaneously, and reduced the risk of detection and punishment. Digital society is therefore particularly criminogenic in terms of fraud — so much so that one could coin the adjective “**fraudogenic**” (Button & Cross, 2017).

Characteristics that make online fraud structurally different from traditional fraud:

- **Scale**: a single offender can target thousands of victims simultaneously.
- **Anonymity**: digital identities are easy to fabricate and difficult to trace.
- **Globalisation**: victims and perpetrators can be in entirely different jurisdictions, complicating prosecution.
- **Low detection risk**: compared to in-person fraud, the likelihood of identification and punishment is significantly reduced.

4.3.2 Fraud Typologies

McGuire and Dowling (2013) identify the following typologies of online fraud:

- **Online frauds and e-commerce frauds**: broad category covering any deceptive scheme conducted via the internet or digital commerce platforms. Includes misrepresentation of goods, non-delivery of purchased items, and payment fraud.
- **Fraudulent sales**: carried out via online auction or retail sites, or through entirely bogus websites created specifically to defraud buyers. The seller collects payment but delivers counterfeit, inferior, or no goods at all.

- **Online mass marketing scams:** individuals are tricked into paying money upfront — for example, to help someone, to participate in a business opportunity, or to claim a supposed prize — with the promise that they will receive a larger sum of money in return at a later date. Classic examples include advance-fee fraud (“419 scams”) and lottery scams.
- **Phishing scams:** a user tricks the victim into providing personal information, financial details, or access codes by posing as a reputable entity or person in an email or other form of communication. The communication typically mimics the branding and language of legitimate institutions (banks, government agencies, e-commerce platforms).
- **Pharming scams:** a user is directed to a fake website — sometimes via phishing emails, sometimes through DNS poisoning — to enter their personal information. Unlike phishing, pharming can redirect users even when they type the correct URL into their browser.
- **Online romance scams:** the victim is contacted via social networks or dating websites and subjected to a prolonged online ‘relationship’, at the end of which they are tricked into handing over personal details or large sums of money. Perpetrators typically fabricate elaborate identities (military personnel stationed abroad, wealthy professionals) and cultivate trust over weeks or months.

4.3.3 Beyond Fraud: Other Economic Cyber-enabled Crimes

- **Intellectual property crimes:** product piracy, counterfeiting, forgery, and the sale of counterfeit goods enabled by global online marketplaces. Digital distribution has dramatically lowered the cost of distributing pirated software, films, music, and physical goods.
- **Sale of illegal items on online marketplaces:** the internet, and particularly the deep and dark web, hosts marketplaces where users can find and purchase weapons, drugs, human organs, stolen data, hacking tools, and a variety of illegal services. The dark web provides layers of anonymity through tools such as Tor that make both buyers and sellers difficult to identify.
- **Cyber laundering:** criminals can clean the proceeds of illegal activities using cyber laundering systems. Some or all stages of the classical money laundering process (placement, layering, integration) may take place in cyberspace — through cryptocurrency transactions, online gambling platforms, or fictitious e-commerce sales — in order to erase the criminal origin of assets.

4.4 Cyber-enabled Crimes Related to Interpersonal Violence

4.4.1 Malicious and Offensive Communications

- **Cyber-bullying:** intimidation and bullying of young people (less often children) through ICTs, including the creation of special websites with material about the person being persecuted, or by posting offensive reviews and comments. Cyber-bullying can be carried out by individuals or by coordinated groups and often extends into the physical world, affecting the victim’s school and social life.
- **Cyber-mobbing:** psychological violence in the form of bullying that uses electronic communication to insult, threaten, or humiliate a victim. Unlike cyber-bullying (which tends to target peers in school settings), cyber-mobbing is typically carried out by a group of people who collectively bully one of their group members or any user, often in workplace or community contexts.

4.4.2 Online Violence Against Other Individuals

- **Cyber-stalking:** the systematic pursuit of a person, a group of people, or an organisation via the internet with the aim of intimidating them, blackmailing them, and making demands. It involves repeated, targeted conduct that causes fear or distress in the victim. Digital tools allow stalkers to monitor victims' movements, contacts, and routines at a level of detail previously impossible.
- **Cyber-harassment:** a repeated psychological cyberattack through persistent insults, harassment, demands, and virtual aggression. It typically manifests as a continuous flow of numerous messages, intrusive calls around the clock, and conversations of a humiliating and insulting nature. The persistence and visibility of online harassment can amplify harm well beyond what equivalent in-person conduct would cause.
- **Image-based abuse (IBA):** a continuum of different forms of sexual violence that includes:
 - Non-consensual intimate image (NCII) distribution
 - Revenge porn
 - Deepfakes or fake pornography (AI-generated sexual imagery using real individuals' likenesses)
 - Sextortion (the use or threatened use of intimate images to coerce victims into further sexual acts or payments)

4.4.3 Online Sexual Offences Against Children

- **Online grooming:** a form of sexual abuse of children and commercial sexual exploitation in which adult users employ digital platforms to establish trust with underage victims, with the ultimate intention of luring them into sexual contact or exploitation. Grooming typically unfolds over time, with the offender progressively desensitising the victim to sexual content.
- **Production and distribution of child pornography:** online platforms, including the dark web, are used to produce, share, and trade child sexual abuse material (CSAM). The digital environment has enabled the formation of large, international networks of offenders who share material and coordinate offending.

4.4.4 Other Forms of Cyber-enabled Interpersonal Crime

- **Trolling:** social provocation in which virtual conversation partners are deliberately drawn into controversy in order to stir up a scandal or inflict emotional distress in network communication. Trolling ranges from low-level harassment to coordinated campaigns targeting individuals.
- **Griefing:** the deliberate infliction of moral and material damage on participants in online computer games — sabotaging gameplay, destroying items, or psychologically tormenting other players. While often considered a minor infraction, in some cases it shades into extortion or fraud.
- **Assisted cyber suicide:** a form of psychological abuse in which the victim is driven to suicide by psychological manipulation, sustained pressure, or encouragement on the internet. It sits at the intersection of cyber-harassment and criminal facilitation of suicide.

4.5 Case Studies

4.5.1 E-commerce Fraud: Belgian VAT Scheme (March 2023)

In March 2023, Operation “Silk Road” uncovered a large-scale VAT fraud scheme involving Chinese e-commerce exporters. The perpetrators evaded Belgian import taxes by falsely declaring that consumer goods (electronics, toys, accessories) were destined for other EU Member States, thereby qualifying for a VAT exemption. The scheme relied on a network of bogus companies registered in France, Germany, Hungary, Italy, Poland, and Spain, supported by false invoices, forged transport documents, and fictitious identities. The total amount of VAT fraud was estimated at **303 million euros**. The case illustrates how cross-border digital trade creates systematic vulnerabilities in tax enforcement regimes, and how international criminal networks can exploit regulatory gaps between Member States.

4.5.2 Cyber Laundering: Estonian Cryptocurrency Fraud (November 2022)

In November 2022, two Estonian nationals were arrested in Tallinn for their involvement in a multi-layered fraud and money laundering scheme centred on cryptocurrencies. The perpetrators operated on two fronts: they tricked victims into signing fraudulent equipment rental contracts, and separately induced victims to invest in a fictitious virtual currency bank. In total, victims paid more than **\$575 million** to the defendants’ companies. The proceeds were then laundered through shell companies and used to purchase real estate and luxury cars in Estonia. The case demonstrates the intersection of classic investment fraud with cyber laundering, and the use of cryptocurrency’s opacity to obscure illicit financial flows.

4.5.3 Online Romance Fraud (March 2023, Italy)

In March 2023, Italian police arrested eight individuals for serious fraud, money laundering, and counterfeiting, following an investigation into a romance scam network. Victims (primarily women) were contacted on Facebook by a fake profile named “Larry Brooks”, a self-proclaimed US Army officer stationed in Syria. Over the period 2018–2021, the criminal group defrauded **32 victims** of approximately **€400,000** in total. The operation relied on a well-established and adaptable modus operandi, with the script adjusted according to the victim’s profile. The case exemplifies the psychological sophistication of modern romance fraud: victims, often elderly or isolated, transferred their savings — including pension payments — in the belief they were supporting a loved one.

4.5.4 Sextortion: The Jordan DeMay Case (March 2022)

In March 2022, Jordan DeMay, a 17-year-old American high school football player from Michigan, was contacted via an Instagram account (“dani.robertts”) that appeared to belong to a young woman. The account had in fact been hacked and sold to a 22-year-old Nigerian man, who used it to coerce young men into sending explicit images of themselves, then threatened to distribute the images unless money was paid. DeMay died by suicide within hours of being targeted. The suspects, Samuel and Samson Ogochi, were subsequently extradited to the United States and charged in federal court. The case became a landmark in debates about the criminal liability for sextortion-induced suicide and the cross-jurisdictional challenges of prosecuting online sexual coercion.

The DeMay case exemplifies the link between IBA and assisted cyber suicide: sextortion does not merely constitute a financial or reputational harm, but in the most severe cases constitutes a precipitating cause of death. It also illustrates how the perpetration of sextortion does not require the offender to possess any intimate images in advance — the scheme is engineered to create them.

4.5.5 Online Sexual Crimes Against Children: The Love Zone

The Love Zone (TLZ) was one of the largest pedophile online communities in the world, with approximately **45,000 members**. The platform was used to disseminate pedophile practices and to sell, procure, and trade child sexual abuse material (CSAM). In 2014, Queensland Police identified the TLZ administrator, covertly assumed control of his account, and monitored the site for six months, exposing the forum's most active and prolific members in a classic law enforcement “honeypot” operation. The operation ultimately led to dozens of arrests across multiple jurisdictions; the most recent arrests linked to TLZ date to May and August 2023. One individual associated with the network was charged with 1,623 child abuse offences. The case highlights the tension between the investigative value of covertly operating illegal platforms (allowing broader identification of offenders) and the ethical and legal implications of temporarily permitting ongoing criminal activity.

CHAPTER 5

Cybercrime Typologies: Cyber-assisted Crimes

5.1 Definition

Cyber-assisted crime is the third and least technology-dependent category in the Brenner–Wall trichotomy. These are traditional crimes in which the internet is used “to assist with the organisation of a crime” (Wall, 2005). The term *cyber-assisted crime* was formally proposed by McGuire and Dowling (2013).

Definition 5.1 (Cyber-assisted crime). Traditional crimes in which the digital tool is **not critical**: the crime “would still take place if the internet was removed” (Wall, 2017). Technology is part of more complex offences or criminal activities, used to **assist pre-existing operations** or by moving part of the criminal business online (Wall, 2017; Di Nicola, 2022). The result is what Wall (2017) calls **cyber-assisted forms of organisation**.

The key distinction from cyber-enabled crime is one of degree, not of kind: in cyber-assisted crime, digital tools intervene at specific phases of a criminal activity that is structurally independent of them. Remove the internet and the crime still happens; with the internet, it becomes more efficient, more scalable, and harder to disrupt.

Remark. The three categories — cyber-dependent, cyber-enabled, cyber-assisted — form a spectrum of digital integration, from crimes that cannot exist without technology, to crimes that merely use technology as an incidental organisational tool. Most real-world organised crime today operates somewhere along this spectrum rather than at a single fixed point.

5.2 Crime Scripts

5.2.1 The Concept

To understand cyber-assisted crime, it is useful to adopt the analytical framework of **crime scripts** (Cornish, 1994). A crime script is a structured description of the sequential steps involved in committing a crime, identifying the actions taken, the resources required, and the decision points at which intervention is possible. Applied to cyber-assisted crime, crime scripts allow us to map exactly *where* and *how* digital tools intervene in an otherwise traditional criminal process.

Key features of the crime script approach:

- Crime is treated as a process that can be “unfolded into separate but related phases” (Lavorgna, 2013).
- The influence of the digital society may be limited to **one or more phases** of the criminal activity, not necessarily the whole.
- The method was recommended by Cornish and Clarke (2002) specifically for investigating organised crime.

- The goal is to understand the role of the internet, the opportunities it creates, and the phases in which those opportunities occur — providing law enforcement with **intervention points**.

5.2.2 Example: Tag Writing (Cornish, 1994)

The classic illustrative example of a crime script is graffiti tag writing. Each stage has a corresponding script action and a corresponding situational control measure that could prevent or disrupt it:

Scene/Function	Script Action	Situational Control
Preparation	Buy spray-can; find a good setting	Sales regulation; city paint-out program
Entry	Enter setting	Access control; entry/exit screening
Pre-condition	Loiter	Surveillance
Instrumental pre-condition	Select target	Remove target
Instrumental initiation	Approach target	Surveillance
Instrumental actualization	Reach target	Protective screens; legal target provided
Doing	Spray graffiti	Graffiti-resistant paint
Post-condition	Get away quietly	Moisture-activated alarm
Exit	Leave setting	Entry/exit screening
Doing (later)	“Getting up”	Rapid cleaning

The value of the script is not the description of any individual act but the systematic identification of *all* the stages at which prevention could interrupt the sequence. The same logic applies to complex organised crimes.

5.2.3 Application: Wildlife Trafficking (Lavorgna, 2013)

Wildlife trafficking is a paradigmatic cyber-assisted crime: a multi-stage smuggling operation encompassing numerous activities (Wyatt, 2022), in which the internet plays a **crucial role specifically in the advertising and selling stages** — making it easier to find potential buyers or sellers (Lavorgna, 2020) — while physical logistics remain offline.

The crime script of wildlife trafficking (Lavorgna, 2013: 78) unfolds across seven stages:

Stage 0 Preparatory activities antecedent to the commission of wildlife trafficking.

Stage 1 Poaching, harvesting, or breeding of the animal or plant.

Stage 2 Intermediate passage through local middlemen and the domestic market.

Stage 3 Passage through regional middlemen and international traders.

Stage 4 Intermediate passage through local middlemen and the domestic market (at destination).

Stage 5 Distribution of the animal, plant, or product.

Stage 6 Activities directly consequential or subsequent to the trafficking activity.

Lavorgna’s detailed mapping of internet use across these stages (2013: 79–81) shows that digital tools appear across all stages as **Crime**, **Lifestyle**, and **Network** script categories. Selected examples:

- **Preparation (Stage 0)**: Formation of criminal networks via email; use of dedicated forums to exchange information on potential buyers; use of Google Translate to post announcements in different languages; communication with other members of the network about relevant legislation.

- **Entry (Stage 0, 1, 4):** Discussion with criminal peers to exchange information on potential new buyers; interaction with potential buyers via auction websites, general commercial websites, dedicated commercial websites, dedicated forums.
- **Pre-condition:** Maintenance of contacts with criminal network members (email, Skype); signalling via dedicated websites of recently discovered species that are rare and valuable; obtaining information on specific species (morphology, breeding); sharing information about the location of the animal/plant in its natural environment (dedicated blogs); agreement to encourage local middlemen who hesitate due to fear of prosecution; informing potential clients and middlemen of updated prices.
- **Instrumental pre-condition (Stages 3, 4):** Agreement to exclude local middlemen who hesitate; search for new suppliers via emails and dedicated forums.
- **Instrumental actualization (Stages 3–5):** Advertising via fairs and exhibitions (dedicated websites, forums); forwarding emails from suppliers who can fulfil orders; identifying the location of the animal via GPS and web mapping services (e.g. Google Maps); ordering/reserving specimens on dedicated commercial websites; arranging sales of the specimen to middlemen via international auction websites; discussing prices and meeting places in the physical world to close deals (email, domains); discussing via criminal network how to ship specimens in small packages with deceptive labels and then send them through regular mail; agreement to physically meet and exchange animals/products (private messaging on dedicated commercial websites, email); putting customers directly in contact with middlemen or suppliers; contacting customers to reassure them that, without documents, the price is significantly lower; discussing with potential customers the absence of required documentation; reassuring customers about the quality of the animal/plant/product on dedicated commercial websites and auction websites.
- **Doing (Stages 1, 4, and later):** Auctioning/selling the animal/plant/product (dedicated commercial websites); selling on auction websites (e.g. eBay); agreement to send animals/plants/products via regular mail among middlemen; monitoring delivery via regular mail; payment via prepaid cards that can be recharged online.
- **Post-condition (Stages 4–6):** Buying accessories and sharing videos and pictures in zoos or cases where wild animals are involved (dedicated forums, deep web); attempting to cover the criminal act by claiming that animals/plants/products have been imported in the past with CITES certificates but have been given to others (dedicated website).

This detailed mapping reveals that the internet penetrates every stage of the wildlife trafficking script, yet the crime itself — the physical capture, transport, and delivery of animals — remains independent of digital infrastructure. This is the hallmark of cyber-assisted crime.

5.3 New Criminal Opportunities

Modern technological tools have created **new criminal opportunities** for organised groups (Lavorgna, 2013, 2015a, 2015b). Understanding this dimension is essential for grasping the full impact of digital society on criminal activities, organised or otherwise.

Lavorgna identifies **five types of opportunities** available to criminal groups:

1. **Opportunities for communication:** encrypted messaging platforms, dedicated forums, and anonymous communication tools allow criminal groups to coordinate at scale and across borders with greatly reduced risk of interception.
2. **Management opportunities:** digital tools facilitate the logistics of criminal operations — inventory management, financial transactions, coordination of supply chains — making

large and geographically dispersed operations manageable.

3. **Organisational opportunities:** the internet enables new forms of criminal organisation, including loosely networked structures that are more resilient to law enforcement disruption than traditional hierarchical groups. Members may never meet physically.
4. **Relational opportunities:** online platforms allow criminal groups to identify, vet, and recruit new members, build trust relationships with suppliers and buyers, and maintain ongoing relationships with co-offenders and clients across jurisdictions.
5. **Promotional, marketing, persuasive, and fidelization opportunities:** criminal groups use the internet to advertise services and products to potential customers, build a reputation, manage reviews and feedback (as seen in dark web marketplaces), and retain clients through loyalty mechanisms analogous to legitimate e-commerce.

5.4 Cyber-assisted Criminal Activities

Cyber-assisted crimes are divided into two broad clusters based on who commits them:

5.4.1 Trafficking Activities of Collectivities

These are multi-actor, multi-stage operations in which organised criminal groups use digital tools to assist pre-existing trafficking activities:

- **Counterfeit medicine trafficking:** the internet facilitates the sourcing, advertising, and distribution of falsified pharmaceutical products, as explored in Project FAKECARE.
- **Smuggling of migrants and asylum seekers:** as documented in the Surf and Sound project, social media platforms and messaging apps are used at every stage of the smuggling process — recruitment, route planning, real-time navigation, and feedback.
- **Trafficking for the purpose of exploitation** (particularly sexual): digital recruitment tools, fake job advertisements, and loverboy methods are used to identify and groom victims before physical exploitation occurs.
- **Terrorism and radicalisation:** the internet is used as a recruitment and propaganda tool, for coordination of attacks, and for dissemination of extremist ideology — while the physical acts of violence remain independent of digital infrastructure.
- **Sale of counterfeit products or psychotropic substances:** online marketplaces (surface web, deep web, dark web) facilitate supply, demand, and payment for a wide range of illicit goods.
- **Illegal animal trafficking:** as illustrated by the wildlife crime script above.
- **Tobacco trade:** illicit tobacco trafficking uses online platforms for sourcing, ordering, and coordinating distribution networks.

5.4.2 Criminal Activities of Individuals

Beyond organised crime, individual criminals also exploit digital tools in an assistive capacity:

- **Using the internet to learn how to commit or hide a crime:** online tutorials, forums, and dark web resources provide information on criminal techniques, from document forgery to evading forensic detection.
- **Using social media accounts to gather useful information about prospective victims:** OSINT (open-source intelligence) techniques applied to social media allow

individual criminals to profile targets, identify vulnerabilities, and plan offences with a level of detail previously available only to professional investigators.

- **Crime-as-a-service:** the digital market allows individuals to outsource specific criminal tasks — hacking, money mule recruitment, document forgery, phishing kit deployment — without possessing the skills themselves, lowering the barrier to entry for complex criminal activity.

5.5 Case Studies

5.5.1 Encrypted Criminal Communications: EncroChat (2020)

EncroChat was an encrypted communication tool used on a large scale by organised criminal groups for criminal purposes. With an estimated **60,000 users**, the platform provided hardware-modified phones running a hardened operating system with features specifically designed to frustrate law enforcement: remote wipe capability, no GPS logging, and end-to-end encrypted messaging. In 2020, a joint investigation by French and Dutch authorities infiltrated the EncroChat network and was able to intercept, share, and analyse millions of communications in real time. The operation led to **over 6,500 arrests** across multiple countries and the seizure of close to **EUR 900 million**. Similar operations subsequently dismantled two further encrypted criminal communication networks: **Exclu** and **Sky ECC**.

The EncroChat case exemplifies the “opportunities for communication” that digital tools provide to organised crime. It also illustrates the law enforcement response: rather than blocking the communications, authorities chose to covertly monitor them, harvesting intelligence on criminal networks at an unprecedented scale. This raises significant questions about admissibility of evidence, bulk surveillance, and the rule of law that continue to be litigated in courts across Europe.

5.5.2 Human Trafficking and Technology

Technology plays an increasingly important role in trafficking in human beings (THB), particularly in the two key phases of the trafficking process: **recruitment** and **exploitation** of victims.

Recruitment phase.

- **Fake job advertisements:** attractive work conditions (high pay, foreign travel, hospitality work) are advertised on legitimate job platforms and social networks, luring victims into exploitative situations before the deception becomes apparent.
- **Loverboy method:** the trafficker cultivates a fake romantic relationship with the victim online, building emotional dependence before introducing conditions of exploitation. This method is particularly effective against young and socially isolated individuals.

Exploitation phase. Once recruited, digital tools are used to control victims (e.g. confiscating phones, monitoring communications), advertise victims’ services online, collect payments, and launder proceeds.

The GRETA report (*Online and technology-facilitated trafficking in human beings*) documents the systematic integration of digital tools into THB operations and the corresponding need for law enforcement responses adapted to the digital environment. A 2023 hackathon specifically targeted the problem of human traffickers luring victims online, bringing together technologists and law enforcement to develop detection tools.

5.5.3 Illicit Online Markets

Digital technologies are used to facilitate the sale of a wide range of illegal and contraband goods. Key categories include:

- **Counterfeit products:** online platforms (including social media and dark web marketplaces) are used to advertise, sell, and ship counterfeit goods. An international operation by Europol shut down 12,526 websites offering counterfeit goods and pirated content, seizing 127,365 counterfeit products worth EUR 3.8 million.
- **Illegal dark web markets:** dedicated marketplaces on the dark web provide a full-service environment for buying and selling drugs, weapons, documents, and other contraband. Law enforcement on three continents arrested 288 dark web vendors in a major marketplace seizure, seizing EUR 50.8 million in cash and virtual currencies, 850 kg of drugs, and 117 firearms.
- **Wildlife and endangered species:** as analysed through the crime script framework above.
- **Cultural heritage artifacts:** the internet facilitates the illicit trade in looted or stolen cultural heritage objects. An international art trafficking operation led to 60 arrests and the recovery of over 11,000 objects.
- **Pharmaceuticals:** online markets, including end-to-end encrypted messaging platforms such as Telegram, are used to distribute both controlled and uncontrolled substances. A documented case involved an underground abortion pill market growing on Telegram as US states restricted access to abortion care, illustrating how legal and political context shapes the demand side of illicit online markets.

Across all three case studies, a common pattern emerges: the criminal activity itself (trafficking, violence, physical delivery of goods) is not digital, but the internet provides the infrastructure for communication, recruitment, advertising, payment, and logistics. This is the operational core of cyber-assisted crime. Law enforcement responses must therefore target not only the physical crime but also the digital layers that enable it — and doing so raises complex questions about privacy, jurisdiction, and the proportionality of surveillance.

CHAPTER 6

AI, ML, and Measuring Cybercrime

Lecture A — AI and ML and Cybercrime

6.1 Human–Machine Interaction and Crime

The literature on digital sociology emphasises the **relationship between the human being and the machine in the construction of the new digital society**, in a socio-technical vision. In a world where robots support or replace humans, and where artificial intelligence is gaining ground, a fundamental criminological question arises: are there, or will there be, increasingly **hybrid criminals** — the result of close interaction between humans and computers?

King et al. (2020: 91) define phenomena of “**crime perpetrated by artificial intelligence**” in which the perpetrator is very difficult to circumscribe, and in which some questions arise spontaneously: *Who commits the crime? A human agent? An artificial agent? Both of them?* These questions challenge the anthropocentric assumptions at the core of traditional criminology and criminal law.

6.2 Artificial Intelligence: The Concept

Definition 6.1 (Artificial intelligence). Artificial intelligence deals with **artificial agents capable of carrying out tasks that usually require intelligence and human intervention to be conducted**. These artificial agents are “sufficiently informed”, “intelligent”, “autonomous and capable of carrying out morally relevant actions independently of the humans who created them” (Floridi and Sanders, 2004: 351).

The key criminological implication of this definition is the concept of **moral autonomy**: an AI system that can take actions independently of its creators creates a category of harmful conduct that sits outside the traditional offender–victim dyad. This forces criminology to reckon with questions of agency, responsibility, and attribution that existing frameworks were not designed to handle.

6.3 AI and Criminology: Three Categories of Challenge

AI introduces new challenges for crime and criminology along three axes:

1. **Crime with artificial intelligence**
2. **Crime against artificial intelligence**
3. **Crime through artificial intelligence**

6.3.1 Crime with Artificial Intelligence

AI can be a **powerful tool for criminal purposes**. As Brundage et al. (2018: 19–22) argue, AI in criminal hands is capable of:

- **Expanding existing threats:** allowing criminals to scale attacks that were previously limited by human labour (e.g. automated phishing at scale).
- **Introducing entirely new threats:** creating categories of harm that were not previously possible.
- **Completely altering the characteristics of existing threats:** changing the speed, accuracy, cost, or detectability of a known crime.

The paradigmatic example is the **deepfake**: AI-generated audio-visual content that realistically impersonates a real person. Deepfakes can be used for disinformation (fabricating statements by public figures), financial fraud (voice cloning to impersonate executives), and image-based sexual abuse (generating non-consensual intimate imagery using a real person’s likeness). As noted in Part IV with the Crosetto case (2025), deepfake technology has already been operationally deployed in large-scale fraud.

6.3.2 Crime against Artificial Intelligence

Here we are dealing with **attacks that exploit the vulnerabilities of an AI system to manipulate or “hypnotise” the system itself**. The AI system is the victim of the attack, but the ultimate objective is typically to weaponise the system against third parties.

Key attack vectors include:

- **Training data poisoning:** the attacker corrupts the data used to train an AI model, causing the model to learn systematically distorted patterns. If the poisoned model is then deployed in a criminal justice, medical, or financial context, its decisions will be systematically biased in ways that benefit the attacker.
- **Adversarial inputs:** carefully crafted inputs that cause an AI system to misclassify or misbehave at inference time — for example, fooling an image classifier or a facial recognition system.
- **Model reprogramming:** manipulating an AI system to commit criminal acts, behaving as an autonomous criminal agent.

The paradigmatic example is **Microsoft’s Tay chatbot** (2016): a conversational AI deployed on Twitter that was manipulated within 24 hours through coordinated user input, causing it to generate racist and offensive content at scale. This is a case where coordinated social engineering constituted a form of “crime against AI” that turned the platform’s own technology into a vector of harm — with no clear individual perpetrator.

6.3.3 Crime through Artificial Intelligence

In this category AI becomes an **intermediary agent**: a non-human actor that stands between the human criminal and the criminal act, creating a layer of separation that complicates attribution and detection.

The clearest example is a **bot purchasing drugs on the dark web** instead of a natural person. The human criminal programmes the bot with intent and provides resources, but the actual criminal transaction is carried out autonomously by the AI. In this model, artificial intelligence

acts as a **criminal shield** — a screen between the offender and the offence that exploits the inability of legal systems to assign criminal liability to non-human agents.

6.4 AI and Crime Areas

AI-facilitated crime is not evenly distributed across crime types. The principal affected areas are:

- **Identity theft and fraud, impersonation, and online fraud generally:** AI enables the creation of synthetic identities, voice and face cloning for impersonation, and automated phishing campaigns personalised at scale.
- **Financial crime:** algorithmic manipulation of markets, AI-driven money laundering through complex transaction patterns, and automated detection evasion.
- **Drug trafficking and trading:** AI bots as purchasing agents, algorithmic pricing and inventory management on dark web markets.
- **Harassment:** AI-generated hate speech, automated pile-ons, and the use of AI to identify and profile harassment targets.
- **Sex crimes:** deepfake pornography, AI-assisted grooming (chatbots maintaining grooming conversations at scale), and automated generation and distribution of child sexual abuse material.

6.5 AI-enabled Future Crimes: Risk Matrix

Caldwell, Andrews, Tanay et al. (*Crime Science*, 9:14, 2020) systematically assessed AI-enabled future crimes along two dimensions:

- **Harm or profit potential** (low to high)
- **Difficulty of defeating** (easier to harder)

The resulting risk matrix classifies AI-enabled crimes as follows:

High risk (high harm, hard to defeat):

- Audio/video impersonation (deepfakes)
- Tailored phishing (AI-personalised social engineering)
- AI-authored fake news
- Disrupting AI-controlled systems
- Large-scale blackmail
- Market bombing

Medium risk:

- Learning-based cyber attacks
- Tricking face recognition systems
- Online eviction
- Data poisoning

Low risk (low harm or easier to defeat):

- Bias exploitation
- AI-authored fake reviews
- Burglar bots
- AI-assisted stalking
- Forgery
- Evading AI detection

The matrix is useful for prioritising defensive investment: high-risk threats that are also difficult to defeat require urgent attention, while lower-risk or more easily defeated threats can be addressed with proportionally lower resources.

6.6 Cyborg-Crime

Working on botnets — networks of infected computers controlled by a single operator, increasingly used to commit cybercrime — van der Wagen and Pieters (2015) coined the concept of **cyborg-crime**: online crime perpetrated by a **cyborg actor**, and proposed the use of **Actor-Network Theory** (ANT) as a framework for interpreting forms of delinquency committed by actors who are *neither completely human, nor completely driven by a machine*.

Their core argument: it is not possible to understand the nature of crimes carried out through botnets if we continue to use an **anthropocentric criminology** that sees the human being as the main force behind criminal behaviour. “Rather than considering a botnet as a technological tool, an individual (opportunistic) crime or a resource on the criminal market, we treat a botnet as a **hybrid crime network**, a crime that derives from mutual cooperation and human/technology interaction” (ibid.: 579).

The cyborg-crime concept bridges the previous three categories (with/against/through AI) and points towards a broader theoretical reconceptualisation of criminal agency in the digital age. It challenges criminology to develop frameworks capable of assigning responsibility, assessing harm, and designing interventions in contexts where the boundary between human and machine agency is systematically blurred.

Lecture B — Measuring Cybercrime: Data Sources and Challenges

6.7 Overview: Crime Data Sources

Traditional crimes can be measured through three main data sources:

1. **Official crime statistics**: quantitative data produced by the criminal justice system.
2. **Victimisation surveys**: data collected directly from the population about their experience of crime.
3. **Self-report surveys**: data collected from individuals about their own criminal or deviant behaviour.

Cybercrime, however, is extremely difficult to measure and to reconcile across these sources. Each instrument faces specific challenges when applied to digital crime, compounding the already severe **dark figure of crime** problem.

6.8 Official Crime Statistics

6.8.1 Definition and History

Definition 6.2 (Official crime statistics). Quantitative data on crime recorded by state agencies, primarily police, prosecutors, and courts. Originated in France in the early 19th century, initially focused on judicial and penitentiary statistics and later extended to police-recorded data.

Main administrative sources at the supranational level:

- **Interpol**: collected comparative data until 2006.
- **UN Crime Trends Survey (CTS)**: collects standardised crime statistics from member states.
- **European Sourcebook of Crime and Criminal Justice Statistics**: compiles cross-national data from European countries.
- **Eurostat database**: produces harmonised EU-level crime statistics.

Key structural problems:

- The **dark figure of crime**: a large proportion of crimes are never reported to police, never recorded, or lost at subsequent stages of the criminal justice process. Official statistics therefore systematically undercount actual crime.
- **No detailed information on the criminal event, the victim, or the offender**: administrative records capture only what is legally and procedurally relevant, stripping out the contextual information needed for criminological analysis.

6.8.2 Official Statistics and Cybercrime

Official statistics on cybercrime are **extremely limited**, for several structural reasons:

- Cybercrimes — especially cyber-enabled ones — are in most cases **registered as traditional crimes**. A phishing fraud is recorded as fraud; a sextortion case may be recorded as blackmail. The digital dimension is lost.
- There is a **lack of uniform registration** of cybercrime in the police registration systems of most Western countries: no agreed taxonomy, no mandatory cybercrime flag, no standardised coding.
- Information concerning the digital nature of a crime may be present in **unstructured qualitative text fields** in police or court registration systems — but extracting it is a labour-intensive task not amenable to standard statistical processing.

Notable exceptions:

- **FBI Internet Crime Complaint Center (IC3)**: a dedicated reporting portal for internet crime in the United States, producing annual reports on cybercrime complaints and losses.
- **Uniform Crime Report Survey of Canada**: includes specific cybercrime categories.

- **European Sourcebook of Crime and Criminal Justice Statistics:** publishes cross-national records of cyber-enabled fraud and, for some countries, cyber-dependent crime.

Methodological response — machine learning: To overcome the problem of cybercrime information buried in unstructured text fields, two research approaches have been developed:

- Sessink (University of Twente): “*Using Machine Learning to Detect ICT in Criminal Court Cases*” — applying NLP classifiers to court documents to identify cases with an ICT dimension.
- Van der Laan and Tollenaar: “*Text Mining for Cybercrime in Registrations of the Dutch Police*” — using text mining to flag cybercrime-relevant entries in police records that would otherwise be invisible to quantitative analysis.

6.9 Victimisation Surveys

6.9.1 Definition

Definition 6.3 (Victimisation survey). A technique that involves asking a representative sample of the population: (1) whether and of what crimes they have been victims in a given period of time prior to the interview; and (2) if so, further details about the incident (where it occurred, at what time, and how).

The key advantage over official statistics is that victimisation surveys capture crimes that were never reported to police, directly addressing the dark figure of crime.

6.9.2 The US National Crime Victimization Survey (NCVS)

The NCVS is the foundational instrument of modern victimisation research. Key milestones:

- **First pilot studies in the 1960s:** conducted on 10,000 people, respondents reported **double the crime rate recorded in official statistics** — a landmark finding that demonstrated the scale of the dark figure and justified the survey approach.
- **Since 1972:** the NCVS has been continuously investigating the frequency, characteristics, and consequences of victimisation in the United States.
- **Coverage:** crimes against the person (rape and sexual assault, robbery, assault, bag snatching/pickpocketing) and against property (theft, burglary, motor vehicle theft).

The NCVS allows researchers to:

1. **Estimate the total number of crimes** committed against the population as a whole — consistently showing a far higher figure than official statistics (the gap between the NCVS and the Uniform Crime Report (UCR) is visually striking, with NCVS figures for violent crime running at several times the UCR count for the same years).
2. **Investigate differential victimisation:** analyse the levels of victimisation to which different segments of the population are exposed, based on victim characteristics (age, gender, ethnicity, marital status, income, education). Example: males, Black individuals, and young people up to age 24 are more at risk of violent crime than women, white individuals, and those over 24.
3. **Provide details about the perpetrator and the events:** capturing information on offender demographics, victim-offender relationship, and circumstances of the offence that official statistics do not record.

Impact of the NCVS: The success of the NCVS prompted the development of similar surveys at both international and national levels:

- **International:** the **International Crime Victims Survey (ICVS)**.
- **National:** Netherlands (Victims of Crime, 1974), United Kingdom (British Crime Survey, 1982), Italy (1997).

6.9.3 Victimization Surveys and Cybercrime

The application of victimisation surveys to cybercrime is still developing. A systematic review by Reep-van den Bergh and Junger (*Crime Science*, 7:5, 2018) — “*Victims of cybercrime in Europe: a review of victim surveys*” — provides the most comprehensive picture to date:

- **Objective:** to map cyber-victimisation patterns in Europe and provide estimates of annual prevalence rates (number of victims per population) for six cybercrimes.
- **Methodology:** definition of survey inclusion criteria; database and internet searches; grey literature; contacts with national statistical offices.
- **Results:** nine surveys identified across European countries; highly heterogeneous annual prevalence rates depending on the crime type:
 - 1–3% for online shopping scams.
 - 3% of the population is a victim of cyberbullying.

The wide variation in estimates across surveys — arising from differences in operationalisation, reference periods, sampling frames, and question wording — underlines how immature the field of cybercrime measurement still is. Comparability across surveys remains a major methodological challenge.

6.10 Self-Report Surveys

6.10.1 Definition and Purpose

Self-report surveys are instruments in which respondents are asked about instances in which they have been **actively involved in crime or deviant behaviour** — i.e., as offenders rather than victims. They are particularly important for:

- Measuring **juvenile delinquency**, where official statistics are especially unreliable due to selective enforcement and diversion.
- Developing and testing **theories of criminal and deviant behaviour**, since they provide data on the prevalence and frequency of offending in the general population rather than only among those caught by the system.

The key international self-report instrument is the **International Self-Report Delinquency Study (ISRD)**.

6.10.2 Self-Report Surveys and Cybercrime

A growing body of self-report surveys incorporates cybercrime measures. Three key instruments (Buil-Gil, Trajtenberg and Aebi, 2025) are:

Name	Acronym	Country	Frequency	Round	Sample
Offending, Crime and Justice Survey	OCJS	UK	Annual, 2003–2006	2006	5,354 respondents aged 10–29
Youth Delinquency Survey	YDS	Netherlands	Quinquennial	2015	1,471 respondents aged 12–23
International Self-Report Delinquency Study	ISRD4	Cross-national	1991/91, 2006/08, 2012/19, 2021/22	2021/22	Varies by country (aged 13–17)

Measures of cybercrime offending used across these surveys (Table 6, Buil-Gil et al., 2025):

- **Malware:** OCJS — “used internet to send viruses on purpose to other computers”; YDS — “intentionally sent out viruses through e-mail or over the internet”.
- **Hacking:** OCJS — “used internet to hack into other computers? by hacking we mean using a computer to illegally get access to another computer’s files”; YDS — “logged on to somebody else’s computer, email or social media account without informed consent”; ISRD4 — “hacked or broken into a private account or computer to acquire data, get control of an account, or destroy data”.
- **DoS:** YDS — “tried to disrupt a website or email account by sending out large amounts of data”.
- **Digital piracy:** OCJS — “used internet to download software, music or films that you knew to be pirated or unauthorised”.
- **Online banking/shopping fraud:** OCJS — “bought anything over the internet using payment card or card details that did not belong to you, without the card owner’s permission”; YDS — “sold something through internet, but not sending out goods after receiving payment”; ISRD4 — “used internet, e-mail or social media to dupe others (like phishing, selling worthless or illegal things, etc.) to make money”.
- **Hate speech:** ISRD4 — “sent hurtful messages or comments on social media about someone’s race, ethnicity or nationality, religion, gender identity, etc.”.
- **Online harassment:** OCJS — “sent email message to someone in order to harass, scare or threaten them”; YDS — “threatened someone through text messages, e-mails or in chat boxes”.

A recurring observation across all three measurement instruments: the dark figure of cybercrime is almost certainly larger than for traditional crime. Victims do not recognise they have been victimised; reporting rates are extremely low; official categories do not map onto cybercrime; and offenders who commit cybercrime often do not perceive their behaviour as criminal. Measuring the true extent of cybercrime remains one of the most significant methodological challenges in contemporary criminology.

CHAPTER 7

Patterns of Online Repeat Victimisation (Correia, 2020)

Reference

Sara Giro Correia, *Patterns of online repeat victimisation and implications for crime prevention*, 2020 APWG Symposium on Electronic Crime Research (eCrime), IEEE. HRC School of Law, Swansea University. Data source: crime reports made by individuals to Action Fraud (AF) within three Welsh police forces (Dyfed/Powys, Gwent, South Wales), reference period 1 October 2014 – 30 September 2016, $n = 10,001$.

7.1 Introduction and Context

7.1.1 Scope of Fraud and Computer Misuse

Fraud and Computer Misuse (F&CM) became part of yearly crime estimates produced by the UK Office for National Statistics (ONS) only in 2015, on the basis of the Crime Survey for England and Wales (CSEW). Recent estimates (year ending March 2020) reported approximately **4.6 million** incidents of fraud and **876,000** incidents of computer misuse, raising the overall crime estimate from about 5.7 million (without F&CM) to **10.2 million** estimated crimes. Only a small fraction of these crimes is reported to police: about **8%** of frauds and **2%** of computer misuse incidents.

Despite low reporting rates, the absolute volume of recorded F&CM has grown considerably, in part because of the rollout of **Action Fraud (AF)** as the National Reporting Centre. Following its national rollout in 2013, the volume of recorded frauds increased by over **160%** between 2011 and 2013. Victim support services have been decentralised, with competitive bidding for providers funded through the Police and Crime Commissioners (PCCs).

7.1.2 Attrition and the ‘Four Ps’ Policing Strategy

There is considerable *attrition* within the criminal justice system: a small share of AF calls are recorded as crimes (‘crimed’), fewer are reviewed for investigation, and fewer still reach courts. Most F&CM victims therefore receive no meaningful CJS response. The UK *Code of Practice for Victims of Crime* requires that vulnerable and repeat victims be identified and offered adequate support, but research on F&CM repeat victims is limited and definitions of vulnerability are inconsistent across agencies.

The ‘Four Ps’ approach

F&CM policing strategy in the UK has four strands, originally drawn from counter-terrorism policing:

- **Pursue** — investigate crimes to enable prosecution.
- **Prevent** — prevent people from engaging in F&CM (awareness of consequences, diversion of talented individuals into legal/ethical hacking).
- **Protect** — increase protection for those at risk of (further) victimisation.
- **Prepare** — improve resilience to reduce the impact of F&CM in the future.

The paper’s contribution falls within the **Protect** strand: targeting those already victimised to prevent re-victimisation.

7.2 Defining and Measuring Repeat Victimisation

7.2.1 Terminology

Several terms describe the experience of being a crime victim more than once: *revictimisation*, *multiple victimisation*, *repeat victimisation*, *multi-victimisation*, *repetitive victimisation*, *recidivist victimisation*.

Definition 7.1 (Repeat victim (CSEW)). Someone who was the victim of **the same crime type more than once** in the previous 12 months (*within crime type victim*).

Definition 7.2 (Multiple victim (CSEW)). Someone who experienced **more than one crime in the previous 12 months, regardless of crime type** (*across crime type victim*).

Definition 7.3 (Repeat victim (this study)). Because the analysis is restricted to F&CM, a **repeat victim** is any individual who reported, within the reference period, more than one fraud, more than one computer misuse incident, or a mix of fraud and computer misuse.

7.2.2 Incidence, Prevalence and Crime Concentration

The extent of repeat victimisation can be gauged from CSEW-derived rates:

- **Incidence Rate (IR)**: estimated average number of incidents per 1,000 respondents.
- **Prevalence Rate (PR)**: estimated percentage of respondents who were victimised (counts victims, not incidents).
- **Crime Concentration (CC)**: ratio IR/PR — average number of victimisations per victim.

For most crime types, repeat and multiple victimisation imply $IR > PR$ and therefore $CC > 1$.

Table I — IR, PR, CC (CSEW, year ending September 2019)

Crime Category	IR	PR	CC
Theft from the person	0.88	0.84	1.05
Robbery	0.32	0.30	1.07
Other theft of personal property	1.44	1.32	1.09
Fraud	8.08	6.94	1.17
Computer Misuse (CM)	2.14	1.82	1.18
Fraud & CM	10.23	8.48	1.21
Violence	2.74	1.76	1.55

The concentration of F&CM is lower than for violent crime but **higher than for comparable acquisitive offences** (theft from the person, theft of personal property), motivating an empirical study of RV with respect to F&CM.

7.3 Patterns of Repeat Victimization in Prior Research

The author summarises four insights from earlier RV literature, used to generate the study's hypotheses.

7.3.1 A Small Share of Victims Carries Most of the Crime

Early work showed that as much as **14%** of the population were repeat victims and accounted for **70.9%** of all incidents recorded on the British Crime Survey. The average number of crimes experienced by the top 10% most victimised households rose from **57% in 1994** to **72% in 2012**, despite the overall drop in crime since the mid-1990s. Reducing RV can therefore prevent a large proportion of all offences from being committed.

7.3.2 RV Does Not Happen by Chance

Failed attempts to fit the distribution of RV to variations of the Poisson model indicate that RV is non-random and is associated with specific demographic characteristics. The characteristics that distinguish repeat from one-time victims tend to be the same as those that distinguish victims from non-victims (e.g., females being significantly more likely to be both one-time and repeat victims of domestic abuse).

7.3.3 The 'Flag' versus 'Boost' Conceptualisation of Vulnerability

- **Flag account:** previous victimisation *indicates* pre-existing vulnerability factors that made the person a victim in the first place.
- **Boost account:** being victimised *itself* increases vulnerability to further victimisation.

In the RV literature 'vulnerability' is often equated with a higher probability of victimisation. Identifying victims as 'vulnerable' on the basis of demographic characteristics alone is divisive and yields too many false positives to be useful in practice. Identifying **repeat victims** is a narrower and more precise indicator of relative vulnerability.

7.3.4 Hot Spots, Hot Dots and Near-Repeats

A key finding from environmental criminology is that high-crime areas ('hot spots') are typically distinguished not by more individuals being victimised, but by **more victims being repeatedly**

victimised ('hot dots'). This has motivated 'near-repeat' research, which predicts where crime will occur next on the basis of where it has previously occurred.

Limitation for F&CM: the spatial element underpinning near-repeats is largely irrelevant for crimes with a strong online component. Even when offline elements are involved (e.g., cold-calling combined with online banking takeover), the physical environment plays only a marginal role in victim selection. Therefore RV is likely a more useful concentration metric than near-repeats for F&CM.

7.3.5 Time-Course

Across burglary, property crime, racial attacks, domestic violence, and computer misoffences, RV has been shown to occur **relatively soon after the first victimisation**: the risk of RV is highest immediately after victimisation.

7.4 Method

7.4.1 Sample

The data are crime reports made to Action Fraud by victims within four Welsh police forces (Dyfed/Powys, Gwent, North Wales, South Wales) between **1 October 2014 and 30 September 2016**. After de-duplication ($n = 147$ duplicates removed), the analytic sample was $n = 17,049$. Of these, $n = 11,844$ pertained to individual victims (rather than businesses or other corporate entities), divided into eight three-month periods (four quarters per year).

For the RV analysis, only reports from **three of the four forces** (Dyfed/Powys, Gwent, South Wales) were used: $n = 10,001$. North Wales was excluded because of high missingness in linkage variables, which skewed the linkage results.

7.4.2 Collection, Linkage and Anonymisation

Reports were collated from the monthly extracts uploaded by the National Fraud Intelligence Bureau (NFIB) onto the police's NicheRMS system and extracted by the Southern Wales Regional Organised Crime Unit (SW-ROCU). Data were processed under strict access controls within secure police premises; **de-duplicative data linkage was applied before anonymisation**, which is what made repeat-victim identification possible.

The reports were linked using a combination of **deterministic and probabilistic** record linkage, implemented with the R package **RecordLinkage**. Linkage quality was tested using **precision** and **recall (sensitivity)** on a clerical review of 100 pairs of matches; both were estimated at 100%. The procedure was tuned to minimise false positives and may therefore have produced false negatives (missed matches).

7.4.3 Limitations of Action Fraud Data

- **Under-reporting of F&CM** (only 8% of fraud and 2% of CM are reported to police).
- **Home Office Counting Rules (HOCR)** affect how RV is measured: in some categories, several incidents over time count as one crime (e.g., advance-fee fraud where the victim sends money to the fraudster repeatedly), while in others (e.g., hacking) each compromised account counts as a separate crime.
- **Left and right censoring**: the two-year window means earlier or later events that would constitute repeats may fall outside the observation period.

- **External events:** an Action Fraud institutional crisis between Q4 of 2014–2015 and Q1 of 2015–2016 sharply reduced the volume of recorded reports.

7.5 Results

7.5.1 Extent of Repeat Victimization

Of the $n = 10,001$ incidents reported by individual victims in the three Welsh forces, linkage identified **350 individual repeat victims**, who reported **787 crimes** between them. Therefore, around **8% of crimes** reported by individual victims (and recorded) were attributable to repeat victims.

Table II — Reports by Crime Group

F = number of fraud reports; CM = number of computer misuse reports; $I\%$ = share of incidents; $V\%$ = share of victims.

reports	F	I%(F)	V%(F)	CM	I%(CM)	V%(CM)
1	8527	92.72	96.53	687	85.45	93.99
2	533	5.80	3.02	59	7.34	4.04
3	88	0.96	0.33	20	2.49	0.91
4	23	0.25	0.07	9	1.12	0.31
5	5	0.05	0.01	20	2.49	0.55
6	21	0.23	0.04	9	1.12	0.21
total ($n > 1$)	670	7.29	3.47	117	14.56	6.02

- Approximately **3% of fraud victims** reported **7% of recorded frauds**.
- Approximately **6% of CM victims** reported **15% of CM crimes**.
- The difference between fraud and CM is statistically significant ($\chi^2(1) = 52.86$, $p < 0.01$); the odds of a CM victim being a repeat victim are **2.17** times those of a fraud victim (small-to-medium effect).

This difference is plausibly driven by HOCR rather than by a genuinely higher rate of RV for CM.

Differences across Welsh forces. A statistically significant difference exists in the proportion of repeat versus one-time reports across the three forces ($\chi^2(2) = 16.26$, $p < 0.01$), but the effect size is negligible (Cramér’s $V = 0.04$).

Table III — Reports by Force and Victim Type

Force	Victim Type	N	St. Residuals
Dyfed/Powys	Once	2210	−3.60**
	Repeat	234	3.60**
Gwent	Once	2200	−0.57
	Repeat	195	0.57
South Wales	Once	4804	3.58**
	Repeat	358	−3.58**

** statistically significant at $p < 0.01$.

The standardised residuals show that reports in **Dyfed/Powys** are significantly more likely to come from repeat victims, while those in **South Wales** are significantly more likely to come from one-time victims.

7.5.2 Repeat Victim Characteristics

Gender

More incidents were reported by male repeat victims than by female repeat victims. The difference is statistically significant ($\chi^2(1) = 14.05$, $p < 0.01$) but the effect size is negligible (Cramér's $V = 0.03$). The odds of an incident being reported by a male repeat victim are only **1.33** times those of an incident reported by a female victim.

Ethnicity

No significant difference was found between one-time and repeat victims in terms of ethnicity ($\chi^2(1) = 0.067$, $p > 0.5$). This result is constrained by **34% missing values** for ethnicity and by the need to collapse to a binary White vs. Black and Minority Ethnic (BME) classification (small numbers in non-White categories).

Age (the largest effect)

Repeat victims are typically **older** than one-time victims:

- Repeat victims: median age **57**, mean age **53.6**.
- One-time victims: median age **50**, mean age **49.93**.
- Wilcoxon rank-sum test: $W = 3,197,004$, $p < 0.001$.

The author notes that linkage may have introduced a bias towards older victims, but date of birth (DOB) was a key linkage variable, and proxy reports on behalf of older victims (by family/friends) would, if anything, increase missingness/inaccuracy of DOB for older victims. The findings are therefore taken as indicative of **a real overlap between RV and older age**.

A broader view of vulnerability

The negative impacts of F&CM on older victims (wellbeing, family relationships, indirect financial loss) may be associated with factors beyond the crime itself, such as social isolation or economic disadvantage. **Vulnerability should be conceptualised more broadly than the increased likelihood of (re-)victimisation alone.**

Proxy reports

It had been hypothesised that reports made by a proxy on behalf of the victim would be associated with greater vulnerability and therefore with RV. The data show the **opposite**:

- **11%** of one-time victims were flagged as having a proxy report.
- **6%** of repeat victims had a proxy report.
- Difference: $\chi^2(1) = 13.58$, $p < 0.01$, Cramér's $V = 0.04$ (negligible).

This is consistent with proxy reports being associated with broader vulnerability factors that are not captured by RV measures alone.

7.5.3 Time-Course of Repeat Victimisation

The time-course analysis is limited to consecutive reports within 12 months ($n = 406$), excluding the small minority of cases ($n = 31$) with a time difference greater than 366 days.

Inter-report time

The distribution of the time elapsed between consecutive incidents is concentrated at the lower end of the scale, with the appearance of an **exponential decrease**:

- **15.5%** of all linked (repeat) incidents were reported **on the same day** ($n = 122$).
- Including same-day reports: mean inter-report difference = 83 days, median = 12 days.
- Excluding same-day reports: mean = 115 days, median = 40 days.
- Mean difference between first and last report by an individual repeat victim = 137 days, median = 49 days (excluding same-day).

Crime prevention implication

Interventions aimed at protecting victims from re-victimisation should ideally take place **within two weeks to one month of the first report**.

The exponential pattern holds across the quarters of the two-year period, although dispersion grows in year two. This is consistent with **left-censoring** (earlier in the window, fewer prior events can be observed) and with the AF crisis between Q4 (2014–2015) and Q1 (2015–2016) that depressed the volume of recorded reports.

Table IV — Time difference between consecutive reports, per quarter

Year	Quarter	n	Mean (diff)	Median (diff)	Max (diff)
2014–2015	Q1	43	4.02	0	45
	Q2	73	12.16	2	136
	Q3	85	34.48	7	208
	Q4	46	43.43	5	306
2015–2016	Q1	23	134.22	122	338
	Q2	45	142.71	97	363
	Q3	36	81.61	25.5	357
	Q4	55	67.33	30	338

Number of repeats

A Kruskal–Wallis rank-sum test confirms a significant overall difference in mean inter-report time across groups defined by the number of reports made ($\chi^2(5) = 1369.8$, $p < 0.01$). Pairwise comparisons (`pairwise.wilcox.test`) indicate that only the group with $n_{\text{reports}} = 5$ has a mean significantly different from the groups with 2–4 reports ($p < 0.05$). Combined with the high dispersion of the data, this suggests that **the number of repeat reports has little practical impact on the time-course of RV**.

Table V — Time difference grouped by number of reports made

n_{reports}	Mean (diff)	Median (diff)	n
2	93 days	35 days	296
3	77 days	45 days	72
4	79 days	83 days	24
5	17 days	28 days	20
6	39 days	28 days	25

7.5.4 Crime Categories and Sequences

Differences in RV across crime categories

Differences in the extent of RV across crime categories are statistically significant ($\chi^2(1) = 52.86$, $p < 0.01$, Cramér's $V = 0.11$, small effect). Standardised residuals indicate that:

- **Advance-fee fraud** and **Hacking**: more likely to be reported by **repeat** victims.
- **Consumer fraud** and **Other fraud**: more likely to be reported by **one-time** victims.

Table VI — Repeat vs one-time reports by crime category

Crime Category	Victim Type	N	Prop	St. Residuals
Advance-fee fraud	Once	190	0.21	-2.95**
	Repeat	7	0.25	2.95**
Card and Banking fraud	Once	445	0.05	2.43
	Repeat	23	0.03	-2.43
Consumer fraud	Once	427	0.46	2.85**
	Repeat	9	0.41	-2.85**
Hacking	Once	439	0.05	-9.76**
	Repeat	102	0.13	9.76**
Investment fraud	Once	174	0.02	-1.76
	Repeat	22	0.03	1.76
Malware, virus, (D)DoS	Once	248	0.03	1.32
	Repeat	15	0.02	-1.32
Other fraud	Once	148	0.16	4.19**
	Repeat	0	0.10	-4.19**
Services fraud	Once	242	0.03	-0.07
	Repeat	21	0.03	0.07

** residual statistically significant at $p < 0.01$.

Sequence of crimes (change matrix)

To reduce small-cell problems, several categories were combined: Malware/Virus/(D)DoS was merged with Hacking; Other and Services fraud were collapsed into Computer Misuse (CM); Card and Banking fraud was combined with Investment fraud.

Table VII — Change matrix of consecutive reports (count, row %, st. residuals)

From\To	Adv.-fee	Card,Bank & Inv.	Consumer	Hacking	Other
Advance-fee	66 58.93% 7.54**	7 6.25% 0.13	23 20.54% -2.9**	3 2.68% -3.8**	13 11.61% -0.9
Card, Bank & Inv.	8 33.33% 0.93	9 37.50% 6.34**	2 8.33% -2.32	1 4.17% -1.59	4 16.67% 0.23
Consumer	21 11.73% -3.36**	6 3.35% -1.42	125 69.83% 7.13**	15 8.38% -3	12 6.70% -2.83**
CM	3 4.35% -3.33**	1 1.45% -1.53	6 8.70% -3.89**	57 82.61% 12.73**	2 2.90% -2.58
Other	7 13.21% -1.61	3 5.66% -0.09	7 13.21% -2.87**	2 3.77% -2.43	34 64.15% 9.3**

** residual statistically significant at $p < 0.01$.

The values along the main diagonal are large, and a chi-squared test confirms a significant pattern in consecutive-crime combinations ($\chi^2(16) = 511.20$, $p < 0.01$, Cramér's $V = 0.54$).

Key result. A repeat F&CM crime is significantly more likely to be **of the same general type** as the crime that preceded it.

7.6 Discussion and Implications for Crime Prevention

7.6.1 Headline Findings

- Roughly **4% of victims report 8% of crimes**; considered separately, **3% of fraud victims report 7% of recorded frauds** and **6% of CM victims report 15% of CM crimes**. The fraud–CM gap is plausibly an artefact of HO�R.
- Categories with elevated RV: **advance-fee fraud** and **hacking**. This is especially striking for advance-fee fraud, since HO�R favour recording multiple fraudster-driven payments as a single crime — in spite of which RV is still observed.
- A statistically significant force-level difference is driven by **Dyfed/Powys**, which has a higher proportion of repeat victims and an older typical victim profile (mean age 55, median 57 in Dyfed/Powys vs. 50 and 48 in Gwent and South Wales).

7.6.2 Counting Rules and the Direction of Bias

Recording practices may simultaneously **over-estimate** RV for CM and **under-estimate** it for fraud:

- **CM (e.g., hacking)**: each compromised account/device is recorded as a separate crime, even if the incidents happen on the same day.

- **Fraud (e.g., romance / advance-fee):** the modus operandi often involves building trust, escalating intimacy and repeatedly requesting money. Multiple incidents are commonly recorded together as one crime.

Even when victims report multiple crimes on the same day, there is still an opportunity to identify them as repeat victims and direct support accordingly.

7.6.3 Profile of the Repeat F&CM Victim

The analysis suggests **male** and **older** victims are at higher risk of repeat F&CM victimisation, while ethnicity and proxy reports show no significant association (within the limits of the data). This profile is different from typical repeat victims of violent crime and domestic violence, but is in line with the general F&CM victim profile, supporting the view that *the characteristics distinguishing repeat from one-time victims tend to mirror those distinguishing victims from non-victims*.

7.6.4 Time-Course and the Two-Week–One-Month Window

In line with prior research on burglary, racial attacks, domestic violence and computer misuse, repeat F&CM victimisation occurs relatively soon after first victimisation. Crime-prevention activity is therefore most effective **within two weeks to one month** of the first report. The scope for intervention is, however, reduced by the **15.5%** of repeats recorded on the same day as the preceding incident.

7.6.5 The Limits of an RV Lens

- RV measurement is shaped by definitions and recording practices; both can over- or under-estimate the phenomenon.
- RV is a **narrower concept than vulnerability**. Whether a victim is re-victimised says little about their support needs or their ability to recover from victimisation.
- A framework of F&CM vulnerability is needed that goes beyond risk of re-victimisation and accounts for victims' wider needs.

7.7 Conclusion

Repeat victimisation is a complex phenomenon affected by under-reporting, recording rules and methodological constraints in linkage. Even so, this analysis demonstrates that identifying and analysing patterns of RV in reported crime produces insights of **theoretical** interest and of **practical** value to the planning and targeting of crime-prevention activities.

- Identifying RV could direct limited resources towards areas where demand for crime prevention and victim support is greatest, both *across* forces and *within* forces (towards crime types where re-victimisation is most common and victims most vulnerable).
- In a context where only **15%** of recorded crimes were 'actioned' in any way (reviewed by NFIB and, where applicable, referred to a police force or partner agency for response), focusing on prevention and protecting victims from re-victimisation is possibly the most effective crime-reduction strategy available.
- Identifying and responding to repeat victims of F&CM may also provide a clear and meaningful measure of law-enforcement performance.

7.8 Summary

- In a two-year sample of $n = 10,001$ AF reports from three Welsh forces, **350 repeat victims** ($\approx 4\%$) accounted for **787 crimes** ($\approx 8\%$ of recorded individual-victim crimes).
- Approximately 3% of fraud victims reported 7% of frauds; approximately 6% of CM victims reported 15% of CM crimes; CM victims have $\approx 2.17\times$ the odds of being repeat victims, plausibly an HOCR artefact.
- **Older age** is the strongest correlate of repeat F&CM victimisation; gender shows a small but significant male skew; ethnicity shows no significant association; proxy reports are *inversely* associated with RV.
- The inter-report time distribution is approximately exponential (median 12 days including same-day reports, 40 days excluding); **15.5%** of repeats occur on the same day; intervention is most effective within **two weeks to one month** of the first report.
- **Advance-fee fraud** and **hacking** show elevated RV; consumer and other fraud are more typical of one-time victims.
- Repeat F&CM crimes are significantly more likely to be of the **same general type** as the preceding crime ($\chi^2(16) = 511.20$, $p < 0.01$, Cramér's $V = 0.54$).
- RV is a useful but **narrower** concept than vulnerability; an F&CM-specific vulnerability framework is needed to address victims' wider needs.

PART II

Profiling, Victimisation, and Prevention

CHAPTER 8

Profiling Cybercriminals

8.1 Who Are Cybercriminals? Findings from the Scientific Literature

8.1.1 From Stereotypes to Research Questions

Popular depictions of cybercriminals revolve around a narrow stereotype: “nerds”—bright but socially inept individuals with very high IQs and advanced technical knowledge, predominantly male, usually teenagers, never violent, and all neatly fitting one profile. Rogers (2011) argues that this image does not survive empirical scrutiny: cybercriminals exhibit **no unique characteristics** that would mark them off as a distinct, homogeneous population.

Once the stereotype is set aside, different studies on the profile of cybercriminals address different research questions, including:

- What is the profile of the *international* cybercriminal?
- Are cybercriminals intellectually superior to traditional criminals?
- What are their motives?

8.1.2 A Framework for Analysis: The Research Process

Profiling studies follow a general empirical-research workflow that frames the studies discussed below:

Choose Topic → Focus Research Question → Design Study → Collect Data → Analyse Data → Interpret Data

8.1.3 Profiling International Cyber Offenders in the US (Hadzhidimova & Payne, 2019)

Hadzhidimova and Payne (2019) focus on the characteristics of **international** cyber offenders prosecuted in the United States.

Methodology. Data drawn from U.S. Department of Justice press releases between January 2009 and December 2017; sample of **225 offenders** who were citizens of a country other than the US; statistical analysis.

Findings.

- Average age: **34.78 years**; **94%** male.
- Nationality: mainly **China (26.7%)**, **Romania (11.6%)**, **Russia (7.1%)**.
- Group offending: more than two-thirds (**68.4%**) of offenders worked in groups.
- Most committed offence types: **cyber fraud** and **hacking**.
- Sentencing: prison sentences predominate; sentence length is significantly higher for African citizens.

8.1.4 Cybercriminals' Motives and Drives

Taylor (1999) — six drives. Addiction, curiosity, boredom, power, recognition, politics.

Rogers (2006). Divides cybercriminals (both cyber-dependent and cyber-enabled) into **eight groups**, integrating criminological typology with motivational analysis.

Brewster (2015) — hackers' motivations. Groups hackers' motives into the following categories:

- Political; ideological; commercial; emotional; informative and promotional; financial; personal; exploitation.

8.1.5 Intellectual Capabilities: Cybercriminals vs. Traditional Criminals vs. Non-Criminals (Schiks, 2022)

Research question. Do cybercriminals have higher intellectual capabilities than traditional criminals or the general population? The study focuses on **cyber-dependent** criminals.

Methodology. Data from Statistics Netherlands; comparison of **CITO test** scores across three matched groups of **143 individuals** each: cybercriminals, traditional criminals, non-criminals.

Results.

- Cybercriminals score **significantly higher** on the CITO test than traditional criminals.
- Cybercriminals score **significantly lower** on the CITO test than non-criminals.

The finding nuances the “intellectually superior” stereotype: cybercriminals outperform other offenders cognitively, but do not exceed the non-criminal baseline.

8.1.6 Cybercriminals and Traditional Criminals over the Life Course (Krananberg, 2018)

Krananberg (2018) analyses **cyber-offending over the life course**, focusing on cyber-dependent crimes. The study examines the extent to which life circumstances in both private and professional life—household composition, employment, education—are differentially related to whether an individual commits cybercrime or traditional crime.

Methodology. Analysis of longitudinal police data and population registration data of all suspects of cybercrime and traditional crime in the Netherlands, 2000–2012.

Household composition effects (similar across cyber- and traditional offending).

- **Cohabiting** makes an individual **less likely** to commit cybercrime than when they live alone.
- Compared to living alone, **single-parenthood increases** the likelihood of committing both cybercrime and traditional crime. Interpretation: having a partner in the household acts as a **source of informal social control**.

Employment and education. Not statistically significantly related to cyber-offending, whereas they reduce the likelihood of traditional offending. The protective effect of work and study against crime does not extend to cybercrime.

8.1.7 The Hybrid Criminal (Hayward & Maas, 2021)

The figure of the **hybrid criminal** emerges from the interaction between human being and machine. Hayward and Maas (2021) propose three categories that re-frame the AI-crime relationship:

- Crimes **with** artificial intelligence (AI as tool);
- Crimes **against** AI (AI as target);
- Crimes **by** AI (AI as autonomous agent).

8.2 Methodological Issues

Bada and Nurse (2021) provide a meta-analysis of the existing literature on cybercriminal profiling and identify the following methodological shortcomings, which explain why robust studies on the profile of cybercriminals are still lacking:

1. **No common definitions:** profiles are sometimes drawn through a general lens rather than referring to specific types of cybercriminals.
2. **No standard approach** to studying criminal behaviour online: research focuses mainly on individual cases of cybercriminals rather than systematic comparison.
3. **Limited research on personality traits** associated with online criminal behaviour.
4. **No data, or great difficulty in accessing data:** a structural obstacle to large-scale empirical work.
5. **Limited focus** (mainly on hackers); overall, perpetrators of **cyber-assisted crimes** are not studied.

CHAPTER 9

Cyber-Criminal Associations and Cyber-Organised Criminals

9.1 Cyber-Criminal Associations

9.1.1 Cooperation Patterns Among Cybercriminals

Compared to traditional crime, links between perpetrators of **cyber-dependent crimes** are **less likely** to result in the exchange of criminal information and in crimes committed in association. The classical criminological intuition that offenders cluster into stable groups holds only weakly online.

Nevertheless, **subcultural social ties** do exist between hackers, who treat one another as “colleagues” even in the absence of formal criminal associations. Three recurring examples of online criminal subcultures are:

1. the modern **hacker** subculture;
2. the **prostitution client** subculture;
3. the **extremist group** subculture.

When cybercriminal associations do form, they remain comparatively uncommon and arise in four ways (the first two being predominant):

1. Exclusively **offline** social contacts.
2. Offline social contacts **plus** an online forum used for recruitment.
3. **Online forums combined with offline** social contacts for recruitment.
4. Exclusively **online** forums.

9.1.2 Focus on the Violent Extremist Subculture (Gaudette, Scrivens & Venkatesh, 2022)

Gaudette, Scrivens and Venkatesh (2022), *The Role of the Internet in Facilitating Violent Extremism: Insights from Former Right-Wing Extremists*, address two research questions:

1. What is the link between the Internet and violent extremism?
2. What are the interactions between the on- and offline worlds of violent extremists?

Methodology.

1. Interview guide developed in consultation with stakeholders.
2. Recruitment of **former right-wing extremists** as study participants.

Key results.

- **Exposure:** about **one third** had their first contact with extreme violence (EV) as a result of accessing radical right-wing content online; individuals encountered in the **offline** world also played a key role; **white-power music** was the most popular form of online content; a recurring driver was the wish “to be part of something”.
- **Immersion:** the Internet **facilitates the process of radicalisation** by immersing participants in extreme right-wing content and a network of like-minded individuals; **seasoned members** play a key role.
- **Interactions between on- and offline worlds** occur across three dimensions:
 - *Activities:* forums, chatrooms and social media sites are used to advertise events held in the offline world; digital platforms support recruitment and contacts at the local level.
 - *Identity:* online personas reinforce and project offline group identity.
 - *Security and surveillance:* members manage operational security across both environments.

9.2 Cyber-Organised Crime

9.2.1 Moral Panic or Reality?

A central debate is whether cyber-organised crime is a documented phenomenon or a case of **moral panic**. The following observations frame the discussion:

- Public and policy narratives often assume a **convergence between organised crime (OC) and cybercrime** without strong empirical evidence.
- Many cybercrime acts require a high degree of **organisation and specialisation**, so the level of involvement of OC groups in cybercrime is likely to be substantial.
- **McGuire (2012)**, based on a large sample of cases, found that **up to 80%** of cybercrime could be the result of organised activity.
- This does **not** imply that the groups involved resemble traditional, hierarchical OC structures, or that they commit **exclusively** digital crime.
- **Wall (2015)** describes cyber-organised crime as having a “**flat**” **structure** and being better characterised as an “**assembly**” rather than an organisation — a *dis-organised* model.

9.2.2 McGuire’s Classification

McGuire proposes a classification of criminal groups that engage in cyber organised crime, structured by the share of online versus offline activity.

- **Type I — Largely online activities**
 - **Swarms** (e.g. “hacktivist” groups): loose, decentralised collectives mobilised around shared causes.
 - **Hubs** (e.g. “scareware” operations): more coordinated, centred on a core that organises distributed activity.
- **Type II — Both online and offline activities**
 - **Clustered hybrids** (e.g. “carding” forums and skimmers): tightly grouped operations combining online forums with offline action.

- **Extended hybrids** (e.g. phone theft and online trading): wider networks bridging multiple physical and digital nodes.
- **Type III** (extension to McGuire’s original two types): groups perpetrating activities that are **predominantly offline** but increasingly intersect with or are mediated through digital environments.

9.2.3 Further Studies: Traditional OC Online (Wang, Su & Wang, 2021)

A complementary research question is: *how do traditional OC groups run their offline business on the Internet?* Wang, Su and Wang (2021), *Organized Crime in Cyberspace: How Traditional Organized Criminal Groups Exploit the Online Peer-to-Peer Lending Market in China* (*British Journal of Criminology*, 61, 303–324), examine this question through the case of traditional Chinese OC groups exploiting online **peer-to-peer (P2P) lending** platforms — an empirical illustration of McGuire’s Type II/III dynamics: offline criminal structures projecting their business model onto a digital marketplace.

CHAPTER 10

Profiling Cyber-Victims

10.1 What Is Victimology?

Definition 10.1 (Victimology). The **scientific study of crime victims**. The term was first used in 1947 by B. Mendelsohn. As a subfield of criminology, victimology studies the **causes/etiology** of victimisation, its **consequences**, and how criminal justice systems and other actors **deal with victims**.

The orientation of the field has shifted over time:

- **Initial focus** on the **role of the victim**: notions of victim precipitation, facilitation, and provocation.
- **1960s** — **more sympathetic view** of crime victims, driven by the victims' rights movements.
- Contributions of the victims' rights movements established victims as bearers of procedural and substantive rights.
- **Victimology today** integrates etiology, consequences, criminal-justice response and victim support.

10.2 From Victimology to Cyber-Victimology

10.2.1 Unique Traits of Cyber-Victimisation

Cybercrime victimisation has features that distinguish it from offline victimisation:

- It **defies the logic of physical proximity**: offender and victim need not share a physical space.
- **Automation** makes it possible to victimise several persons with the same act, generating large numbers of victims at low marginal cost.

The “human” element and harm of cybercrime was recognised only in **2001**, by the **Council of Europe Convention on Cybercrime**.

10.2.2 Typology of Cyber-Victimisation (Jaishankar, 2012/2013)

Jaishankar (2012, 2013) proposes the following classification of cybercrimes by target:

1. Cybervictimisation of **governments**;
2. Cybervictimisation of **corporations**;
3. Cybervictimisation of **individuals** (interpersonal cybervictimisation);
4. **Victimless crimes**.

10.3 Measuring Cybercrime through the Victims' Lens

10.3.1 Victimisation Surveys and Cybercrime (Reep-van den Bergh & Junger, 2018)

Reep-van den Bergh and Junger (2018), *Victims of cybercrime in Europe: a review of victim surveys* (*Crime Science*, 7:5), address the question: *what about cybercrime victimisation surveys?*

Objective. Map existing cyber-victimisation surveys in Europe and provide estimates of **annual prevalence rates** (number of victims per population) for **six cybercrimes**.

Methodology. Definition of inclusion criteria for surveys; search in databases and on the Internet, in the grey literature, and through contacts with national statistical offices.

Results.

- **9 surveys** identified.
- Different annual prevalence rates depending on the crime: e.g. **1–3%** for online shopping fraud; **3%** of the population is a victim of cyberbullying.

10.4 Explaining Cybercrime Victimisation

Four theoretical perspectives are used to explain why some individuals become cyber-victims.

10.4.1 Lifestyle Theory (Hindelang et al., 1978)

Certain behaviours related to **work, school and leisure** expose individuals to risky (crime-prone) situations: *certain behaviours elevate one's odds of being victimised*.

10.4.2 Routine Activity Theory (RAT) (Cohen & Felson, 1979)

Definition 10.2 (Routine Activity Theory). Victimisation results from the convergence in time and space of three elements:

1. a **motivated offender**;
2. a **suitable target/victim**;
3. the **absence of capable guardianship** (humans or objects that can prevent crime from occurring).

RAT *describes* victimisation.

Although Lifestyle Theory and RAT are conceptually distinct, they lead to similar hypotheses and are often combined into the **Lifestyle Routine Activity Theory (LRAT)**.

10.4.3 Cyberlifestyle-Routine Activity Theory (Reyns et al., 2011)

Reyns et al. (2011) revisited LRAT to adapt it to cybercrime, producing the **cyberlifestyle-routine activity theory**, which re-interprets RAT's convergence requirement for the digital environment:

- **Convergence in place:** offender and victim are often physically separate, yet they come together in the same *virtual* space, via computer networks.

- **Convergence in time:** offender and victim may not interact in real time, yet they intersect in time (e.g. the offender sends an email that the victim reads some time later).

10.4.4 General Theory of Crime (Gottfredson & Hirschi, 1990; Schreck, 1999)

The key factor in causing crime and deviance is **low self-control**: the inability of an individual to exercise personal restraint in the face of tempting immediate and easy gratification, both in the short and long term. The theory was developed for *offending* and was **adapted to victimisation** by Schreck in 1999.

10.4.5 Actor-Network Theory (van der Wagen & Pieters, 2018)

Actor-Network Theory (ANT) is invoked to account for emerging forms of **high-tech cybercrime** (such as ransomware, botnets and virtual theft) in which the offender targets a **hybrid victim** — a composite of **human, technical and virtual** entities.

10.5 Who Are Cyber-Victims? Findings from the Scientific Literature

10.5.1 Many Studies, Many Research Questions and Methods

Different studies ask different research questions and use different methods to investigate cyber-victims (e.g. individuals vs. business victimisation; focus on different cybercrime types such as cyberbullying or identity theft; focus groups vs. surveys/interviews). Overall, the amount of research remains **limited** and findings are sometimes **conflicting**.

10.5.2 A Routine-Activities Analysis (Junger et al., 2017)

Aim. Investigate the relationship between users' **routine activities** and **socio-economic characteristics** and three forms of cybercrime victimisation: **online shopping fraud**, **online banking fraud**, and **DDoS**.

Methodology. Data from Eurobarometer (sample of **17,811 European citizens**); logistic regression.

Results.

1. Findings generally **support RAT**.
2. Socio-economic variables (gender, age, economic factors) are **almost completely irrelevant**: the lecture summarises this finding as the **normalisation of cyber-victimisation** — under universal connectivity, exposure ceases to be a function of socio-demographic profile.

10.6 Focusing on Repeat Cyber-Victimisation

10.6.1 Definition and Supertargets

Definition 10.3 (Repeat victimisation). Experiencing more than one victimisation. Pease (2018) states: *“People, places, products or services, vehicle, household, business, or systems (including virtual networks) are, once victimized, more likely to be victimized again. Further victimization by the same type of crime is most likely.”*

Definition 10.4 (Supertargets). The **1–2%** of potential targets that experience around **half** of crime.

10.6.2 Predicting Crime: Space–Time Clustering of Residential Burglaries (UK)

Many studies show that **previous victimisation** is a risk indicator. A line of work in the United Kingdom investigates residential burglaries to address the question: *do residential burglaries cluster in time and space?*

Theoretical basis.

- **Optimal foraging theory:** animals optimise foraging strategies to maximise benefits while minimising risks and search time — the same logic is applied to offenders selecting targets.
- **Transmissibility** (from epidemiology): closeness in space and time of epidemics/diseases. *Does this also apply to home burglary?*

Methodology.

- Georeferenced data on burglaries in **Merseyside (6 years)**.
- Statistical tests to verify whether space–time clustering was a random event or not.
- For each street, calculation of the space–time distance between break-ins.

Results.

- **Time of repeat victimisation** (Johnson, Bowers & Hirschfield, 1997): the distribution of weeks between events is sharply right-skewed — repeats concentrate in the **first weeks** after the initial event and decay over the following year.
- **“Neighbourhood” effect:** in the 1-week window, the number of burglaries peaks at the victimised house and decays with the number of doors apart; the 2-week window shows a flatter but still elevated profile near the original target.
- **“Side of the road” effect:** houses on the **same side** of the road as the victimised house experience more subsequent burglaries than those on the opposite side, controlling for house-number distance.
- **Area at risk:** houses within **400 m** of a victimised home are at increased risk for **up to one month** after the first burglary.
- **Prospective hotspot prediction:** predicts **62%** of robberies that occur in the following **2 days** and **64%** of those that occur in the following **7 days**.

Burglary in a home is **“transmissible”**, like a disease: the immediate neighbourhood of a victimised home is at elevated risk for a bounded time window. This transmissibility logic underpins predictive policing tools that target proximate locations within short post-event windows.

10.6.3 Connection to the BJS and Correia Studies

The repeat-victimisation logic developed in the burglary literature feeds directly into the two readings discussed elsewhere in these notes: the BJS *Repeat Violent Victimization, 2005–14* (Chapter 11) on the offline side, and Correia (2020) *Patterns of Online Repeat Victimization* (Chapter 7) on the online side.

10.7 From Cyber-Victimisation to Prevention: Your Human Firewall

The lecture closes on a prevention frame: cybersecurity ultimately depends on the user. The recommended viewing/discussion item is the TED talk *Your Human Firewall — The Answer to the Cyber Security Problem* by Rob May (TEDxWoking), <https://www.youtube.com/watch?v=BpdcVfq2dB8>, which argues that the most effective layer of defence against cyber-victimisation is the **human** layer — security-aware behaviour by individual users.

CHAPTER 11

Repeat Violent Victimisation (BJS, 2005–2014)

Reference

Bureau of Justice Statistics, *Repeat Violent Victimization, 2005–14*, Summary, NCJ 250567, August 2017. Data source: National Crime Victimization Survey (NCVS), 2005–2014.

11.1 Overview and Definitions

The BJS summary reports on the patterns of repeat nonfatal violent victimisation in the United States during the period 2005–2014, drawing on NCVS data. The reference population is persons aged 12 or older.

Definition 11.1 (Nonfatal violent victimisation). For the purposes of the report, nonfatal violent victimisation comprises four offence categories: **rape or sexual assault, robbery, aggravated assault, and simple assault**.

Definition 11.2 (Repeat violent victimisation). A person experiences repeat violent victimisation when they suffer **two or more** violent victimisations during the year.

Headline figures (2005–2014 average):

- An average of **3.2 million persons** age 12 or older experienced one or more nonfatal violent victimisations during the year.
- **81%** of victims experienced a **single** nonfatal violent victimisation.
- **19%** of victims experienced **repeat** violent victimisation (two or more victimisations during the year).
 - **14%** experienced **two to five** violent victimisations.
 - **5%** experienced **six or more** violent victimisations.

11.2 Concentration of Victimisation

A central finding is that repeat victims, although a minority, account for a disproportionate share of all violent victimisations.

- In **2014**, the **19%** of violent crime victims who were repeat victims accounted for **50%** of all violent victimisations.
- The concentration was even more pronounced at the upper end of the distribution: the **5%** of victims who experienced six or more violent victimisations accounted for **27%** of total violent victimisations in 2014.

Methodological implication: a small subset of highly victimised individuals contributes a very large share of overall crime counts. Crime prevention strategies that target this subset can, in principle, achieve substantial reductions in total victimisation.

11.3 Crime Prevalence Rates: Long-Term Trends

The report compares prevalence rates across two decades (1993 vs. 2014), expressed as the number of victims per 1,000 persons age 12 or older.

- **Total nonfatal violent crime prevalence rate:** decreased **62%**, from **29.3** per 1,000 in 1993 to **11.1** per 1,000 in 2014.
- **Single violent victimisation prevalence rate:** decreased **60%**, from **22.4** per 1,000 in 1993 to **8.9** per 1,000 in 2014.
- **Repeat violent victimisation prevalence rate:** decreased **69%**, from **6.9** per 1,000 in 1993 to **2.1** per 1,000 in 2014.

The decline in repeat victimisation (69%) was steeper than the decline in single victimisation (60%) over the same period.

11.4 Victim–Offender Relationship

Patterns of repeat violent victimisation differ markedly according to the relationship between victim and offender.

11.4.1 Share of Victims Experiencing Repeat Violence, by Relationship

During 2005–2014, the percentage of victims who experienced repeat violent victimisation, broken down by the relationship to the offender, was:

- **Intimate partner violence (IPV) victims: 33%.**
- Victims of violence committed by a **well-known or casual acquaintance: 26%.**
- Victims of violence committed by a **relative: 25%.**
- Victims of violence committed by a **stranger: 17%.**

IPV victims who experienced repeat violent victimisation accounted for more than **two-thirds (68%)** of all violent crime committed against IPV victims — the strongest concentration observed across relationship categories.

11.4.2 Repetition by the Same Type of Offender

Repeat violent crime victims were more likely to experience two or more victimisations by the **same type** of offender than by different types of offenders. Specifically:

- **26%** of IPV victims experienced repeated violence by an intimate partner.
- **19%** of victims of violence committed by a well-known or casual acquaintance were victimised at least twice by a well-known or casual acquaintance.
- **17%** of victims of violence committed by a relative experienced repeat relative violence.
- **11%** of victims of violence committed by a stranger experienced repeat stranger violence.

11.5 Types of Repeat Violence

Repeat victimisation also varies by offence type.

11.5.1 Share of Victims Experiencing Any Repeat Violent Victimization, by Offence

During 2005–2014:

- **Rape or sexual assault:** 31% of victims experienced repeat violent victimisation.
- **Simple assault:** 23%.
- **Aggravated assault:** 21%.
- **Robbery:** 19%.

Rape or sexual assault victims experienced repeat violent victimisation at a higher rate than victims of any of the other three offence categories.

11.5.2 Repetition of the Same Offence Type

A greater percentage of rape or sexual assault victims experienced two or more victimisations of the same kind than was the case for the other offence categories:

- **Rape or sexual assault:** 16% of victims experienced two or more rape or sexual assault victimisations.
- **Aggravated assault:** 9% of victims experienced repeated aggravated assault.
- **Robbery:** 7% of victims experienced repeated robbery.

11.6 Summary

- Most violent crime victims (81%) experience a single victimisation in a given year, but the minority (19%) who experience repeat victimisation account for half of all violent victimisations, and the very small share (5%) experiencing six or more accounts for over a quarter of the total.
- All prevalence rates — total, single, and repeat — declined substantially between 1993 and 2014, with the steepest decline in the repeat victimisation rate (–69%).
- Intimate partner violence shows the highest rate of repeat victimisation across relationship categories (33%), and IPV repeat victims account for 68% of all violence committed against IPV victims.
- Repeat violence tends to come from the **same type** of offender, particularly in IPV cases.
- Among offence types, rape or sexual assault shows the highest rates of both any repeat violent victimisation (31%) and repeat victimisation of the same offence type (16%).

CHAPTER 12

Responding to Cybercrime: Prevention Strategies

12.1 Approaches to Crime Prevention: Primary, Secondary and Tertiary

12.1.1 What Is Crime Prevention?

Definition 12.1 (Crime prevention (United Nations, 2002)). Strategies and measures that seek to reduce the risk of crimes occurring, and their potential harmful effects on individuals and society (including fear of crime), by intervening to influence their multiple causes.

In operational terms, crime prevention covers **any activity** by an individual or group (public or private) which attempts to **eliminate crime prior to it occurring or before any additional activity results**.

The dominant categorisation adopts a **public-health approach**: focus on preventing ill-health by encouraging healthy lifestyles rather than waiting for illness — that is, *investing funds early to save costs later*. By analogy, crime prevention is articulated into three levels:

12.1.2 The Three Levels

Primary prevention (universal). The most universal form of crime prevention, largely indiscriminating and targeting **society as a whole**. Intended to target and **prevent criminal behaviour before it occurs**. Interventions either **reduce crime opportunities** or **enhance social factors** that reduce an individual's likelihood of becoming involved in crime. In what follows, primary prevention is analysed through **Situational Crime Prevention** and **Universal Communication Strategies**.

Secondary prevention (at-risk groups). Targets population groups who have **not yet engaged** in criminal behaviour **but are at risk** of doing so (e.g. kids showing signs of deviant/criminal behaviour). Focus on people with increased proclivity for crime, **before they graduate into more serious offending**. Analysed through **Educational Workshops**, **Mentoring Programs**, and **Targeted Warnings and Cautions**.

Tertiary prevention (known offenders). Targets individuals **after** they have become involved in crime. Focus on activities aimed at preventing those who have already engaged in criminal activity from **future reoffending**. Individuals are formally referred to programming by the criminal justice system following a criminal conviction. Analysed through **Positive Diversions** and **Restorative Justice Practices**.

12.2 Primary Cyber-Crime Prevention

12.2.1 Situational Crime Prevention (SCP)

Definition 12.2 (Situational Crime Prevention (Clarke)). SCP “*comprises **opportunity-reducing measures** (1) directed at highly specific forms of crime, (2) involving the management/design/manipulation of the immediate environment in as specific and permanent way as possible, (3) so as to **increase the effort and risk** of crime and **reduce the rewards** as perceived by a wide range of offenders.*”

Theoretical foundation. The key assumption is that many offenders are **rational** and weigh the costs and benefits of offending. SCP builds on three theoretical strands:

- the **rational choice** offender approach;
- **routine activities theory**;
- **crime pattern theory**.

The 25 SCP techniques. Clarke and colleagues organise SCP interventions in a 5×5 matrix along five mechanisms:

1. **Increase the effort** (e.g. target harden, control access to facilities, screen exits, deflect offenders, control tools/weapons);
2. **Increase the risks** (e.g. extend guardianship, assist natural surveillance, reduce anonymity, utilise place managers, strengthen formal surveillance);
3. **Reduce the rewards** (e.g. conceal targets, remove targets, identify property, disrupt markets, deny benefits);
4. **Reduce provocations** (e.g. reduce frustrations and stress, avoid disputes, reduce emotional arousal, neutralise peer pressure, discourage imitation);
5. **Remove the excuses** (e.g. set rules, post instructions, alert conscience, assist compliance, control drugs and alcohol).

12.2.2 Are SCP Measures Effective in Preventing Cybercrime?

The lecture focuses on five SCP measures translated to the cyber domain:

1. **Antivirus software:** an effective strategy.
2. **Warning messages:** see the factsheet *What works in situational cybercrime prevention?*
3. **Primary Crime Prevention Apps:** many typologies of apps but no rigorous evaluation; see the article *Primary Crime Prevention Apps: A Typology and Scoping Review*.
4. **Internet vigilantism:** civilian users acting as informal guardians online.

12.2.3 Universal Communication Strategies

Mass-media messages (broadcast on TV or other audio/video channels) that aim to **deter people from committing crimes**. Two key primary mechanisms:

1. **Changing perceptions of the risks** of offending.
2. **Promoting positive behaviours and values**.

A special additional category is the **appeal to third parties to intervene**.

Application to cybercrime. Very little has been done to tackle cybercrime through universal communication strategies, with few exceptions. The flagship example is **Cyberchoices** (UK National Crime Agency), <https://nationalcrimeagency.gov.uk/what-we-do/crime-threats/cyber-crime/cyberchoices>.

12.3 Secondary Cyber-Crime Prevention

12.3.1 Educational Workshops

Bring together groups of (potential) offenders in **single or multiple sessions** to educate them about the nature of crime and reinforce prosocial values and behaviours. Features:

- Deployed within school curricula, or selectively target **at-risk adolescents**.
- Combine **didactic and interactive** elements.
- Rely on **social-cognitive theories of learning**.
- Mainly aimed at **youth in primary or secondary schools** to affect **early-onset offending**.

Applied to cybercrime. Minimal empirical research to date.

- **i-SAFE program** (<https://isafeventures.com/>). The 2006 evaluation reported: “*positive and significant changes in knowledge between the treatment and comparison groups, both on average and over time. For the most part, there were no significant changes in behavior between the treatment and comparison groups.*”
- Arizona Attorney General’s **Social Networking Safety Promotion and Cyberbullying Prevention (SNSPCP)**.
- **Media Heroes**.

A recurring pattern: education programs reliably move **knowledge**, but the effect on actual **behaviour** is often null or weak.

12.3.2 Mentoring

Definition 12.3 (Mentoring). “Commitment of time and specific efforts by a **more experienced** person to the development of a mutually beneficial, supportive and nurturing relationship with a **less experienced** person.”

Mentoring offers a **corrective experience and intervention** for at-risk youth who have experienced poor relations with parents or other caregivers. The theoretical underpinnings emphasise the importance of **positive and supportive interpersonal relationships** that offer guidance to children and adolescents throughout their development.

Targets. Young people who either exhibit a history of delinquent behaviour or characteristics assessed as at-risk for juvenile delinquency (individual and environmental risk factors).

Applied to cybercrime. So far **no evaluation research** examining the utility of mentoring specifically for cybercrime; lessons must be drawn from studies on offline crimes.

12.3.3 Targeted Warnings and Police Cautions

Used by police to warn potential offenders with the aim of **detering them from future offending**. Take the form of stand-alone warnings, cautions, reprimands, and instructions to “cease and desist”, which may be delivered by law enforcement officers in person or through letters.

Applied to cybercrime.

- UK investigation into **Blackshades**;
- **LizardStresser** booter service;
- International law-enforcement operations.

All aimed at people **already involved** in cybercrime or at least on the periphery.

12.4 Tertiary Cyber-Crime Prevention

12.4.1 Positive Diversions

Redirect individuals towards **prosocial behaviours and peer influences**; focus on **rehabilitation**. Usually targeted at young people and first-time offenders. Take a variety of forms, including art, sports, and outdoors programs.

Theoretical underpinnings.

- Imposing traditional criminal justice sanctions on young people has **little effect on reoffending**.
- Positive diversions lean on **social-bond theories of crime**.

Operational notes. May serve as a **substitute** for criminal sanctions or court proceedings, or accompany them. Widely deployed in response to various low-severity crimes. Effectiveness shows **mixed findings**.

Applied to cybercrime.

- No empirical evaluation of positive-diversion programming for preventing cybercrime so far.
- It is plausible that one could **redirect the skills used for cybercrime into prosocial avenues** (e.g. cybersecurity).
- A special category transforms criminal into noncriminal behaviour: e.g. a young person engaging in graffiti diverted into a program encouraging legal “urban art”. By analogy, graffiti artists and cybercriminals share a **sense of identity, pleasure from crime, and status culture**.

12.4.2 Restorative Justice

Definition 12.4 (Restorative Justice). A **voluntary, community-based** response to criminal behaviour that attempts to bring together the **victim**, the **offender**, and the **community** in an effort to address the harm caused by the criminal behaviour.

Theoretical link. Strongly linked to **Braithwaite’s theory of re-integrative shaming**.

Process. Can occur **prior to, instead of, during, or after** the formal criminal process. Tends to take the form of **single meetings** of varying duration, chaired by a **mediator**, producing an **agreement** between the parties.

Applied to cybercrime.

- **No studies to date** specifically on restorative justice's application to cybercrime.
- Wealth of evidence in the offline context, with mixed findings.
- Application to cybercrime could help **convey a message of the harmfulness** arising from such crimes; some cybercrime victims have expressed interest.
- Case study: **online hate speech** (Finland) — <https://yle.fi/aihe/artikkeli/2019/11/06/face-to-face-with-a-man-who-wished-an-asylum-seeker-would-kill-my-family-the>.

CHAPTER 13

What Works in Situational Cybercrime Prevention (NSCR Factsheet, 2021)

Reference

Asier Moneva and Jim Schiks, *What Works in Situational Cybercrime Prevention?* NSCR (Nederlands Studiecentrum Criminaliteit en Rechtshandhaving) Factsheet, June 2021.

13.1 Motivation and Scope

For decades, **Situational Crime Prevention (SCP)** techniques have proven effective in reducing various forms of crime across many contexts, with substantial experimental support. Cybercrime researchers have sought to **extend SCP to cyberspace**. Despite the wide variety of SCP techniques, criminologists have so far applied only a few to cybercrime, evaluating:

- the effectiveness of **antivirus software** in detecting malware;
- the ability of **monitoring software** to detect anomalous patterns of activity among employees;
- — especially — the effect of **warning banners** on cybercriminal behaviour.

13.2 Situational Crime Prevention: Recap

SCP consists of a set of **25 techniques** that reduce specific forms of crime through five mechanisms:

1. increase the **effort** required to commit a crime;
2. increase the **risk** of detection;
3. reduce the potential **rewards** of crime;
4. reduce **provocations** to potential offenders;
5. eliminate **excuses** for non-compliance.

These techniques operate **immediately before the crime occurs**, when all previous crime controls (personal and social) have failed. If situational prevention also fails, only formal crime controls (police, criminal justice system) can be invoked — they are costly and sometimes conflict with people's rights, so it is best to avoid them when possible.

13.3 Methodology of the Overview

The factsheet focuses on **research with high validity**: studies that measure cybercrime **before and after** an intervention in experimental and **comparable control** conditions — i.e. studies

scoring **three or more** on the **Maryland Scientific Methods Scale**.

The only studies that meet this criterion are **experiments with warning banners**. All other SCP techniques applied to cybercrime have not yet been evaluated with comparable rigour.

13.4 Experiments with Online Warning Banners

Definition 13.1 (Online warning banner). A small text frame, usually displayed to users when they try to access a system or a website. Banners work like traditional signs: placed in easily visible places to persuade the reader with a clear, short message. Criminologists use them to **increase the perceived risk** for cyber-offenders or to **remove any excuse** for non-compliance.

Experimental design. To measure the effect of banners on human behaviour, researchers use **honeypots**: digital decoys designed to be attacked that collect data on the attack and the behaviour of the offender. Once cyber-offenders fall into the trap, they are randomly assigned to groups exposed to different stimuli — a banner with a text (e.g. “you are being watched” or “this is illegal”), a neutral banner (e.g. “the site is loading”), or no banner at all. This allows comparison of the effect of different stimuli on similar groups of offenders to identify which one works best.

13.5 Do Banners Work?

Experiments with warning banners have mainly targeted three behaviours:

- **Deterring hackers from trespassing** into systems;
- Deterring users from consuming **barely-legal pornography**;
- Deterring **DDoS attacks**.

13.5.1 Hacking / Trespassing

For hacking, experiments measured the effect of banners on four outcomes: number of trespassing events, number of repeated trespassing events, duration of the trespassing, and volume of commands entered into the system.

- **Mixed effect** on the number of trespassing and repeated trespassing events and on the number of commands entered: some banners reduced these behaviours, others had no effect.
- Banners appear to **significantly reduce the duration of the trespassing** — the time spent by the offender inside the system.

13.5.2 Barely-Legal Pornography

The single experiment on this topic found that banners displaying a deterrent message **halved the volume of accesses** to the website hosting barely-legal content.

13.5.3 DDoS Attacks

When banners were used to divert young people from carrying out DDoS attacks, researchers observed a change in the **increasing trend of attacks to a flat trend**.

13.6 Conclusion

Despite the growing interest in tackling cybercrime, experimental research on the effect of SCP techniques on deflecting cyber-offenders **is still in its infancy**. High-validity research has so far evaluated only a few techniques. With limited exceptions, criminologists have concentrated on **increasing the perceived risk** of detection and **removing excuses** for non-compliance — especially through warning banners — to deter hackers from breaking into a system.

It is not yet clear whether banners are always an effective tool for reducing cybercrime, but experiments show **promising results in specific contexts**. The factsheet calls for **more research** both on banners and on the other situational techniques that remain unexamined.

SCP categories already explored with warning banners. *Increase the risks* and *remove the excuses*. The other three mechanisms — *increase the effort*, *reduce the rewards*, and *reduce provocations* — remain largely unexplored in the cybercrime context with rigorous experimental designs.

CHAPTER 14

Primary Crime Prevention Apps: A Typology and Scoping Review (Wood, Ross & Johns, 2021)

Reference

Mark A. Wood, Stuart Ross and Diana Johns, *Primary Crime Prevention Apps: A Typology and Scoping Review, Trauma, Violence, & Abuse*, 2021. SAGE. DOI: 10.1177/1524838020985560.

14.1 Motivation and Aims

Over the last decade, an array of smartphone apps has been designed to prevent crime, violence, and abuse. Their evidence base has not been analysed systematically. The article frames this as part of the “**appification**” of crime prevention — the creation of apps designed to prevent crime through a variety of measures, including crime reporting, personal safety, rehabilitation and reintegration.

The review has three aims:

1. Establish the **extent, range, and nature** of research into smartphone apps with a primary crime-prevention function.
2. Locate **gaps** in the primary crime-prevention app literature.
3. Develop a **typology** of primary crime-prevention apps.

14.2 Theoretical Frame

14.2.1 Brantingham & Faust (1976) — Tripartite Model

The review adopts Brantingham and Faust’s tripartite model, which distinguishes **primary**, **secondary**, and **tertiary** crime prevention. Primary crime prevention apps “*seek to modify or abate the social or environmental conditions that give rise to crime, or allow potential victims to manage their exposure to risky environments or circumstances.*” Secondary and tertiary apps (those that modify offender behaviour or treat conditions accompanying offending) are excluded from this review.

14.2.2 Wall (2007) — Three Levels of Smartphone Involvement

Drawing on Wall’s (2007) distinction between cyber-enabled and cyber-dependent crime, the authors distinguish three relations between a crime-prevention technique and smartphone technology:

- **Smartphone-enabled:** the technique can use smartphones but does not require them, and is not enhanced by being incorporated into one.
- **Smartphone-enhanced:** smartphones improve the speed, range, or effectiveness of the technique.
- **Smartphone-dependent:** the technique depends on mobile affordances and cannot exist without the device.

14.2.3 Mobile Affordances

The reviewed apps draw on two affordances of smartphones:

- **Perpetual contact** (Mascheroni & Vincent, 2016): the ability to communicate with others anywhere and anytime — underpins personal safety and crime-alert apps.
- **Geo-tracking:** real-time locational information and crime-avoidance messaging sensitive to an individual's movements.

14.3 Method

Methodology. Scoping review following the **PRISMA** guidelines. Articles identified via three databases (**Web of Science**, **EBSCOhost**, **Google Scholar**), supplemented by snowballing. Search terms combined “app(s)/smartphone application” with terms such as “access to justice”, “crime”, “crime prevention”, “violence prevention”, “rape prevention”, “justice”, “personal safety”.

Inclusion criteria. (1) published **2008–2020**, (2) examined or detailed smartphone applications designed **explicitly for primary crime prevention**, (3) written in **English**. Inclusion criteria were deliberately broad to capture engineering and computer-science *software proposals* (the design phase), not only evaluations.

Sample.

- Initially sampled: **151 papers**.
- Excluded: 31 (not smartphone-based, etc.); 2 (not designed for crime prevention); 57 (secondary/tertiary apps).
- Final corpus: **61 academic works** (34 peer-reviewed articles, 17 conference papers, 7 book chapters, 3 university research briefings).

Study-design breakdown. Experimental ($n = 3$), quantitative cross-sectional ($n = 12$), qualitative ($n = 9$), mixed method ($n = 2$), software proposals ($n = 22$), conceptual analyses ($n = 12$), scoping reviews ($n = 1$).

Analysis. Inductive–deductive thematic analysis, coded by the three authors plus a research assistant. Articles were assessed along: *app features*, *study design*, *crime-prevention efficacy*, *advantages and disadvantages*.

Limitations acknowledged. (1) Few studies systematically evaluate apps' preventive efficacy — most describe design or critique unintended consequences. (2) The review excluded general social-media apps (Facebook, WhatsApp) sometimes repurposed for crime prevention.

14.4 A Typology of Primary Crime Prevention Apps

The review identifies **six types** of apps targeting potential victims and addressing criminogenic conditions in the physical and social environment.

14.4.1 Self-Tracking Personal Safety Apps ($n = 24$)

Aim to promote personal safety, primarily by reducing the user's risk of being victim of a violent crime. Mechanisms:

1. enabling friends or family members to **track the user's location**;
2. sending an **alert** to a friend, family member or law-enforcement agency in situations where the user feels in danger;
3. automatically sending an alert to family members after a set amount of time **if the user does not personally deactivate** it.

Majority of literature ($n = 14$) concerns apps designed specifically for **women's safety**. Effectiveness mostly unassessed; one effective ($n = 1$), one ineffective ($n = 1$) in formal evaluations.

Advantages. May increase users' feelings of safety in public space; may collect data on factors contributing to fear of crime; "passive" alerts can prompt alarms without requiring intentional action from the user.

Disadvantages.

- May reinforce **victim-blaming** attitudes by focusing on victims rather than offenders' behaviour.
- Apps may become redundant in light of existing smartphone affordances.
- Often fail to address the broader social and institutional context of violence against women, focusing only on individual factors.
- May be unfeasible and a safety-risk for individuals with disabled phone-sharing partners — can be used to facilitate **stalking**.

14.4.2 Child-Tracking Apps ($n = 3$)

Apps designed to promote child safety through **geo-tracking of other individuals** (children and family members). Example: *Life360*.

Concerns.

- May create a situation of over-proximity in which parents mistake control for care.
- May create rather than relieve parents' anxieties.
- May infringe on children's rights to privacy.
- May amplify harmful power dynamics in intimate relationships and be used to facilitate stalking.

14.4.3 Crime-Mapping / Alert Apps ($n = 17$)

Apps that promote personal safety through **crowdmapping crime** and providing users with alert services and information on "safer" routes and locations in real time.

Definition 14.1 (Crowdmapping (Quaintance, 2011)). The aggregation of crowd-generated inputs such as text messages and social-media feeds with geographic data to provide real-time, interactive information on events.

Examples:

- *WikiCrimes* — platform harnessing collective intelligence and anonymous mass collaboration for citizen crime prevention.
- *Walksafe* — campus safety app evaluated against Pennsylvania State University’s emergency notification system.
- *StreetSafe, Hollaback!* — targeted at women to combat street harassment and sexual assault.
- *Sex Offender Tracker* — mapped specific crime types; criticised for relying on unreliable and outdated official registries.

Disadvantages.

- Often rely on subjective and racially-coded perceptions of what constitutes “suspicious” behaviour, leading to profiling of racial and socio-economic groups.
- May fuel **vigilantism**.
- Crime-mapping apps may draw on official databases that are historic, unreliable, or out-of-date.
- Striking case: *SketchFactor* and *GhettoTracker* (renamed *Good Part of Town*) were quickly labelled racist and classist; both were taken down within days.

14.4.4 Decision Aid Apps ($n = 7$)

Focus on preventing **intimate partner violence (IPV)** through personalised information and decision aids that assist individuals in making decisions based on high-quality evidence. Example: *MyPlan Safety Decision App* (and trials such as *Thrive*, *weWomen*, *ourCircle*). Notable feature: app-based interventions can be disisable, addressing privacy concerns raised by survivors.

14.4.5 Crime Reporting Apps ($n = 14$)

Allow users to easily report various forms of crime. Unlike crime-mapping apps, reporting apps do **not** provide access to user-submitted data on reported events. Examples: *Crime Stoppers*, New Orleans’ *Tip Submit*, *Project Jagriti* (crowdsourced child abuse reporting), *Project Spear* (human trafficking), *iWitnessed* (guided-recall to improve witness accounts).

Open issue. Such apps must comply with data-protection legislation (e.g. EU GDPR); Charitou et al. (2018) showed via *TRILLION* that security-by-design can reconcile compliance with operational efficacy.

14.4.6 Education Apps ($n = 4$)

Provide users with **uniform, generic** educational material about recognising specific forms of crime. Unlike decision-aid apps, they do not deliver individually tailored messages or plans. Examples:

- *SAP_MobAPP* — primary-school children, teaches recognition of sexually-abusive behaviour/situations.

- *Thrive* — aimed at mothers who have experienced IPV.
- Mendel & Sharapov’s (2020) content analysis of 63 “anti-trafficking apps” found that **75%** had fewer than 100 downloads, **17%** had non-functional features, and many spread **misinformation** about human trafficking.

Pfaller (2017) / Žižek’s concept of **interpassivity** is invoked: rather than educating users about people trafficking, such apps may produce *different forms of ignorance about it* — “ignorance facilitates an interpassivity where it is believed that the apps are acting in an individual or organization’s place.”

14.5 Discussion

The crime-prevention landscape is shifting: where prevention used to be a mediated, top-down process (police, government, councils), users now choose from a marketplace of mobile apps. Some are developed by traditional crime-prevention providers, others by specialist app developers or individual entrepreneurs. Consumers need informed advice on **what they need** and **how to use it**.

Two key gaps.

1. **Large gaps in product documentation and evidence** for many apps currently available. App stores’ technical and content reviews are limited; inaccurate or low-quality content is common.
2. **Near-absence of systematic evaluations** of app outcomes — especially for “personal safety” apps. No studies to date have rigorously examined the efficacy of women’s personal safety apps, crime-mapping apps, child abuse prevention apps, or generic personal safety apps in meeting their primary aim: *reducing victimisation from crime*.

Limited interdisciplinarity. Only seven articles in the review were published in criminology journals or books; crime prevention app development is substantially driven by software-engineering considerations with limited dialogue between computational and criminological knowledge.

Critical social-science perspective. Social scientists have predominantly concerned themselves with the **unintended consequences** of these apps:

- Apps like *Circle of 6* aim to “*limit or alter the behaviour of potential victims*” rather than perpetrators.
- Crime prevention apps have been read as **responsibilisation** strategies, transferring risk management to individuals (Comack & Peter, 2005; Gray, 2009; Mowlabocus, 2018).
- “Responsibilisation has been *transcoded* into the language and logic of the smartphone” (Manovich, 2001).

14.6 Implications for Practice, Policy and Research

The article’s headline implications:

- To date, **no studies** have evaluated the efficacy of child-abuse prevention apps, crime-tracking/mapping apps, or crime-reporting apps. *Recent enthusiasm over these forms of*

victim-facing app-based crime prevention interventions is not supported by adequate research or evaluation evidence.

- Numerous studies highlight the need for **evaluative research** that considers potential **unintended consequences** of such apps (e.g. increasing users' fear of crime or victim-blaming sentiments).
- Few software proposals ground their designs in **crime-prevention theory** or evidence: greater dialogue needed between computational and criminological knowledge.
- In the absence of systematic effectiveness reviews, users need advice on choosing an app appropriate to their needs and a realistic appraisal of the **likely benefits and risks** (including privacy and data-security risks) of using them.
- Crime-prevention agencies should carry out **systematic user reviews** of apps and provide "**curated**" apps that offer useful prevention assistance without compromising users' safety or security.